

Two Threads Beat One, and the Thread Hop Is a Wash Under Hawkes Arrivals: A Closed-Form and Empirical Case for Multi-Actor HFT Pipelines

Vincent Maciejewski*

Draft v2, September 2026

Abstract

The received wisdom in high-frequency-trading (HFT) system design is that the entire hot path — SBE decode, order-book application, strategy touch, order construction — should live on a single-thread event loop, because every additional thread hop adds real wall-clock latency. This paper shows that received wisdom is wrong on real CME Market Data Platform version 3 (MDP3) packet flow, and reports two contrarian conclusions.

First: the thread-hop cost is a wash. The received single-thread wisdom assumes that a $\sim 1.7 \mu\text{s}$ one-way async mailbox hop (Maciejewski [33] Table 4; the `fast_send` paper’s $3.4 \mu\text{s}$ round-trip halved) is a material addition to end-to-end latency. On real Hawkes-clustered CME MDP3 packet arrivals, the observed cluster-wait tail is one to two orders of magnitude larger than that per-hop cost at every service floor from 4 to $32 \mu\text{s}$, so the hop is not a first-order effect. The single-thread objection — “every hop is a fixed tax that no benefit can overcome” — is empirically wrong on this class of feed.

Second: two threads beat one on the tail at the cost of one hop on the median. Splitting a service task into two half-length actors on separate threads joined by one $1.7 \mu\text{s}$ mailbox hop lowers p_{99} on 100% of 3512 thirty-minute windows across 276 sessions at every service floor T from 4 to $32 \mu\text{s}$, cutting the corpus-median tail excess to 0.28–0.34 of its single-thread value while adding exactly h to the median; below the $2h$ floor, at $T = 2 \mu\text{s}$, it loses, as the design equation predicts. The p_{99} -minimising stage count rises with the task length: 2 at $4 \mu\text{s}$, 4 at $8\text{--}16 \mu\text{s}$, 8 at $32 \mu\text{s}$. The same pattern generalises to an N -stage series tandem: the end-to-end wait is at most $1/N$ of the single-stage wait plus $(N - 1)h$, at every quantile and on every sample path, capped by the number of natural cut points in the decode chain (four or five on CME MDP3).

The effect is entirely arrival-law-driven: under a rate-matched Poisson null on the same arrival count, both wins vanish and the multi-actor architecture strictly loses. Two closed-form results anchor the mechanism (a burst-limit throughput identity, and an exact pathwise reduction of the deterministic-service tandem to a single server that yields the $1/N$ bound); a literature search over foundational tandem theory, Hawkes-adjacent queueing, and empirical HFT/RPC work found no direct prior art on either the theoretical statement or the empirical calibration. Companion code contains the message-tape parser, the per-window packet-Hawkes fitter, and the tandem Lindley simulator.

*M2 Tech (16425640 Canada Inc.), Montreal. mayeski@gmail.com. KASPAR-HFT is open source at <https://github.com/vincent212/kaspar-hft>.

Contents

1	Introduction	4
1.1	The design choice, and the safe default	4
1.2	Prior recommendations for single-thread hot paths	4
1.3	What this paper does	6
1.4	Contribution and prior-art positioning	7
1.5	Note on the use of AI in this paper	7
1.6	Notation	7
I	Analytical framework	9
2	Analytical framework	9
2.1	Burst-limit throughput theorem	9
2.2	Exact reduction of the tandem to one server, and the scaling bound	10
2.3	What the theory says, in plain terms	14
II	Empirical experiment	17
3	Experiment setup	17
3.1	Data corpus	18
3.2	Packet arrivals are Hawkes-clustered (empirical prerequisite)	18
3.3	Simulator	19
3.4	Pipelining is what makes multi-stage work	19
3.5	Sweep design	20
4	Empirical results	21
4.1	Tail collapse under Hawkes, and its absence under Poisson	21
4.2	Median cost is linear in N , exactly as $(N - 1)h$	23
4.3	Design equation	23
4.4	Across windows	25
4.5	Sensitivity to h	25
5	Evaluation: do empirics match the analytical predictions?	26
III	Best practices: implications for HFT systems design	28
6	The reference instance: KASPAR-HFT and its fast_send pipeline	28
7	Implementation on KASPAR-HFT: two dispatch paths, one handler	29
7.1	Receiver transparency	30
7.2	Why measurement, not local reasoning, must decide	30
7.3	Operational discipline	31
7.4	Separation of code construction from latency optimization	31
7.5	Contrast with non-fungible frameworks	32
8	Other architectural considerations to re-evaluate under Hawkes	32

9 Recommendations for HFT system designers	34
9.1 What the paper leaves untouched	34
9.2 What the paper does argue against	34
9.3 Recommended design flow	34
 IV Conclusions	 36
10 Conclusion	36
11 Future work	37
12 Reproducibility	38
 Appendices	 42
A The Borel batch-size tail at fixed branching ratio	42
B Numerical check of Theorem 1	42
C Numerical check of the unequal-stage claims	43

1 Introduction

1.1 The design choice, and the safe default

The design of the ingress-to-order pipeline in a high-frequency trading system is a choice between two archetypes: the single-threaded event loop that processes every step of every message on one core (SBE decode, book application, strategy evaluation, order construction, all under one call chain, taking a total service time we denote T), and the multi-actor pipeline in which those steps are split across N separate actors, each on its own thread with its own mailbox, connected by $N - 1$ explicit hops whose per-hop wall-clock cost we denote h . It is not really a debate: almost everyone in the low-latency HFT community is in the single-thread camp, and single-threaded is generally viewed as the safe choice. The argument for it cites the cost of a mailbox hop — typically a mailbox enqueue plus a cache-line transfer between the sender’s core and the receiver’s core, plus the cost of the receiver being scheduled or spinning — and holds that $(N - 1)h$ adds directly to every message’s end-to-end latency, so every extra stage in the pipeline is a fixed tax that no throughput or tail benefit can overcome.

That argument implicitly assumes an arrival process for which single-server throughput $1/T$ is comfortably larger than the peak arrival rate. On CME MDP3 the arrival process is strongly self-exciting: packets arrive in Hawkes-clustered bursts [24–26]. During such bursts the instantaneous arrival rate transiently exceeds the single-thread event loop’s service rate $1/T$, and cluster-induced queueing accumulates a tail that dwarfs h by one to two orders of magnitude. This paper’s central empirical observation, on 276 sessions (3512 thirty-minute windows) of NQ front-month packet arrivals, is that this cluster tail is what dominates observed end-to-end latency: the 99th-percentile latency (p_{99}) of a constant-service queue driven by real packet arrivals is 3.5 times the service floor at the corpus-median window for an $8\mu\text{s}$ task and 8 times for a $32\mu\text{s}$ task, and 4.4 and 10.7 times at the 95th-percentile window. Under a rate-matched Poisson null of the same arrival count the p_{99} equals the median; the tail is entirely a Hawkes-cluster effect.

Given that the tail is cluster-driven, the natural question is whether splitting the pipeline into $N > 1$ stages — each running as an independent actor on its own core, with per-stage service T/N , so that the tail-driving cluster-wait is now split across N smaller queues — reduces the tail. If it does, and if the additional median cost $(N - 1)h$ is small relative to the tail savings, the multi-actor architecture is the better trade for any tail-weighted operator, and the single-thread default is displaced for HFT workloads specifically. All symbols in this paragraph are gathered in the notation table of Section 1.6.

1.2 Prior recommendations for single-thread hot paths

The single-thread rule has a specific lineage in the low-latency-systems community, and we name that lineage here because the paper is arguing against it directly. In reverse chronological order, the lineage has three layers: the HFT-specific codification of the last decade, the general low-latency-systems and event-driven-server culture from which HFT inherited it, and one prominent published counter-argument that the field discarded.

Modern HFT-community codification (2010–present). The founding statement of the current HFT-community formulation is Thompson’s Single Writer Principle [36] — “a single-writer design consistently outperforms a ‘properly’ lock-free or wait-free multi-writer design on most realistic workloads” — motivated by and applied in the LMAX Exchange architecture [37, 38]. Rigtorp’s widely cited low-latency tuning guide [39] formalises the operational rule (“run a single thread in

SCHED_OTHER per core and use busy waiting/polling”), and Chronicle Software’s public engineering materials [40] adopt the same architecture for market-data connectors (“within a single event loop . . . busy-spinning and affinity locks”), pinning Chronicle Queue’s design to a single writer per queue. The Aeron messaging library [41] inherits the same lineage. The academic side is thinner, but Kwan et al. [42] codify the pattern as the standard low-latency-HFT design pattern, and recent practitioner posts [43] restate it explicitly (“all critical trading logic runs in a single-threaded event loop, eliminating context switches and lock contention entirely”). Inside HFT specifically, the same recommendation has been reinforced by two decades of private prop-trading arms-race practice among the Chicago desks (Getco, Jump, Optiver, DRW; late 2000s), reported second-hand in industry retrospectives but without an open-literature reference.

Older event-driven-server lineage (1995–2010). HFT did not invent this rule; it inherited it from the event-driven-server culture. The single-threaded event loop as an architectural pattern dates at least to Schmidt’s Reactor pattern [44] in the mid-1990s. It was sharpened by Kegel’s C10K essay [45], which named the OS thread-per-connection model as the bottleneck of scalable network servers and argued for event-driven designs. The pattern went mainstream in production with Sysoev’s nginx web server [46], whose single-threaded worker process is explicitly justified as context-switch avoidance, and later with Dahl’s Node.js runtime [47], which pushed the single-threaded event loop into ordinary application code.

Pro-pipelining minority — outside HFT. A legitimate multi-decade pro-pipelining tradition exists in the datacenter and networking-systems literature; it has been the field’s non-single-thread minority throughout the period covered by the previous two paragraphs. The earliest prominent published argument is the Staged Event-Driven Architecture (SEDA) of Welsh et al. [48], which advocated exactly the multi-stage pipeline of independently scheduled event handlers that the present paper’s tandem design revives. Welsh’s 2010 retrospective [49] is often cited as a walk-back to single-thread, but it is actually a hedged pro-pipelining recommendation: it advocates decoupling stages from thread pools, grouping multiple stages within a single thread-pool domain where latency is critical, and putting a separate thread pool and queue in front of any stage with long or nondeterministic runtime — effectively the present paper’s design rule in different vocabulary. Beyond SEDA the pro-pipelining tradition includes the Click Modular Router [50] for staged packet processing; Naiad [51] for low-latency dataflow; Snap [52] for Google’s multi-engine network stack; and the datacenter microsecond-tail-latency scheduler line — Shinjuku [53], Shenango [54], Caladan [55], Perséphone [56] — which argues for centralised multi-core dispatch specifically to control the microsecond-scale tail. Perséphone in particular is targeted at “wide service-time distributions,” the closest existing framing to our arrival-law claim; but the datacenter line frames the problem as heavy-tailed *service*, not clustered *arrivals*. DPDK explicitly supports both run-to-completion and pipeline modes and its documentation recommends pipeline mode “if some set of packets require longer to process” [57]. Inside HFT specifically the pro-pipelining position is rare in the open literature; an obscure Chinese patent [58] describing a pipelined FAST market-data decoder is a lonely commercial artefact of it.

None of the pro-pipelining sources above ties the design recommendation to Hawkes / self-exciting / near-critical arrival dynamics. The closest theoretical hooks are the MMPP-input tandem work of Kim and Chydziński [16] already cited in Section 1.4 and the batch-arrival tandem phase-transition analysis of Vinnikov [59]. The present paper’s arrival-law-specific mechanism — that the recommendation flips when arrivals are near-critically self-exciting — remains uncontested by prior work.

Yours truly gets a footnote on that list, but with an important caveat.¹ An earlier paper by the present author [33] did not argue for the single-thread principle in its own right. That paper is defensive rather than prescriptive: the actor model had long been dismissed as unsuitable for HFT because of asynchronous-dispatch overhead, and `fast_send` is presented there as the mechanism that eliminates the overhead for co-located actors and thereby makes the actor model usable *inside* a servicing chain that the received wisdom of the time had already placed on a single thread. The reference architecture in that paper — reader $\rightarrow$ buffer via asynchronous `send`, decode/book/signal chain via `fast_send` on one thread, signal $\rightarrow$ order-out via asynchronous `send` — takes the collapsed single-thread servicing chain as the widely accepted design context of the time and delivers a way to compose it out of actors at zero context-switch cost. It does not compare single-thread against more-asynchronous topologies and pick single-thread; it takes the topology as given. The present paper is a retraction of that design context, not of the `fast_send` mechanism: on Hawkes-clustered feeds, the aggregate servicing chain should be split into asynchronous stages once its total service time becomes long compared to the Hawkes-cluster tail. `fast_send` remains the right mechanism for chaining actors that have been placed together on one thread; what changes is where the thread boundaries inside the servicing chain should sit. The corrected design rule is stated in Section 10.

1.3 What this paper does

We answer that question along three axes.

- **Theory.** Two closed-form results (Section 2) predict the pipeline behaviour. Proposition 1 shows that under any arrival process an N -stage tandem enjoys an N -factor throughput speedup on isolated bursts. Theorem 1 shows that a deterministic-service tandem reduces exactly to a single server with the largest stage’s service, and that the end-to-end wait is bounded pathwise by $1/N$ of the single-stage wait plus $(N - 1)h$, at every quantile and under every arrival sequence. Corollary 2 shows that under a rate-matched Poisson null at the feed’s utilisation the single-stage wait is already zero, so there is nothing to compress and splitting costs exactly $(N - 1)h$. The Hawkes cluster representation then explains why the single-stage wait on a real feed is two orders of magnitude above T : the Borel cluster size has scale $(1 - n)^{-2}$.
- **Simulation.** We build an N -stage tandem Lindley recursion on real CME MDP3 packet-arrival streams at a hop cost $h = 1.7 \mu\text{s}$ (the one-way async mailbox cost derived by halving the round-trip figure in Table 4 of Maciejewski [33], and consistent with Maciejewski [32]) and a service floor swept across $T \in \{2, 4, 8, 16, 32\} \mu\text{s}$ to cover the regime from tasks below the hop cost (where no split helps) through tasks well above it (where the tail-win compounds). We sweep $N \in \{1, 2, 4, 8\}$ per scenario on each of the 3512 thirty-minute RTH windows of the corpus. Each window is also simulated under a uniform-shuffle Poisson null of the same arrival count as an empirical counterfactual (Section 3.5).
- **Design equation for practitioners.** From the empirical result we extract a stage-count rule in terms of the operator’s tail weight and the largest stage of the partition, and work it

¹A related author’s-aside: the tail compression reported by this paper had in fact been observed empirically in production well before its mechanism was understood. Splitting a long task into shorter asynchronously connected stages was known to help the tail on a per-strategy basis in the author’s own operational experience, and the working assumption at the time was that the single-thread camp must simply be writing faster inner-loop code — not that the architectural recommendation was itself arrival-law-dependent. The present paper is the retrospective explanation of that anecdote.

through on the KASPAR-HFT actor chain that carries the SHADOW-POV execution algorithm [35] (Sections 6 and 7).

1.4 Contribution and prior-art positioning

Two rounds of literature search covering foundational tandem theory (Jackson, Kelly, Burke, Kleinrock, Whitt), tandem theory under bursty arrivals (Foss–Korshunov, Baccelli–Foss–Lelarge, Heindl, Kim–Chydzinski, Dębicki–Mandjes), Hawkes single-queue and adjacent (Daw–Pender, Chen–Blanchet, Koops–Boxma–Mandjes, Sheldon et al., Gao–Zhu, Bo Li, and 2023–2026 follow-ons), and empirical/applied HFT and RPC-pipeline work (ABIDES, ABIDES-MARL, JAX-LOB, Aquilina–Budish–O’Neill, Dean–Barroso, Parsimon, and the datacenter tail-latency crowd) confirmed that there is no direct prior art on either (i) the theoretical claim that the tandem-vs-single-queue architecture recommendation flips sign as a function of arrival law (Poisson vs Hawkes) or (ii) an empirical CME-calibrated tandem sweep with a Poisson null and measured HFT-framework hop cost. The full reference list appears in Section 12.

1.5 Note on the use of AI in this paper

This paper was produced with Anthropic’s Claude (via Claude Code) used under the author’s direction for code generation — the message-tape parser, the packet-Hawkes fitter, and the tandem-Lindley simulator run on the 3512-window corpus — and for exploratory analysis, mathematical drafting, prior-art literature search, prose editing, and figure preparation. The analytical results of Part I, including the proof of Theorem 1, were drafted with the model and then independently checked by the author. One episode is worth recording: an earlier model-drafted heavy-tail conjecture (an $N^{-1/2}$ tail-ratio built from the Borel–Tanner cluster tail and an $M^{[K]}/D/1$ single-big-batch principle) was found on review to be unsound — the Borel law is light-tailed at every $n < 1$ (Appendix A) — and was replaced by the present, stronger reduction theorem. The author independently reviewed all mathematical arguments, experimental methodology, numerical results, and citations, and takes sole responsibility for the contents of the paper.

1.6 Notation

The following symbols are used throughout.

Symbol	Meaning
T	total service time per message on a single-stage pipeline, in μs
N	number of stages in the tandem pipeline (one actor per stage)
T/N	per-stage deterministic service time in the N -stage tandem
h	per-hop wall-clock cost of the async mailbox handoff between adjacent stages
k	size of an arrival burst (number of packets in one cluster)
$\tau_k^{(N)}$	exit time of the last message of a k -burst under N stages
$s_i, s_{\max}$	per-stage services of an unequal tandem, $\sum_i s_i = T$, and their maximum
$w_j(s)$	Lindley waiting time of message j at a single deterministic server with service s
$W_j^{(N)}, W_q^{(N)}$	end-to-end wait of message j (latency minus T , hops included) under N stages, and its q -quantile
p_q	q -th percentile of the end-to-end latency distribution (p_{50}, p_{99} , etc.)
$\Delta(N)$	clustering tail excess $p_{99}(N) - p_{50}(N)$ (§4.3)
$\Delta_{\text{single}}(s)$	single-server clustering tail excess at deterministic service s ; $\Delta(N) = \Delta_{\text{single}}(T/N)$ (§4.3)
γ	fitted exponent in $\Delta(N) \approx \Delta(1)N^{-\gamma}$
κ	operator's tail-versus-median weight in the design equation
N^*	stage count minimising the design objective $(N-1)h + \kappa \Delta(N)$ (§4.3)
μ, α, β	Hawkes background rate, jump size, and decay rate (§3.2)
$n = \alpha/\beta$	Hawkes branching ratio
K	Borel-distributed offspring cluster size (§2.2)
$\bar{\lambda}_{\text{pkt}}$	mean packet-arrival intensity per 30-min window, packets per second
n_{pkt}	branching ratio of the packet-arrival Hawkes fit per window
$n_{\text{pkt}}^{\#}$	per-window packet count
ρ	utilisation, $\rho = \bar{\lambda}_{\text{pkt}} \cdot T$
$F(T)$	Fano factor at window length T : $\text{Var}[C_T]/\text{E}[C_T]$
H	Hurst exponent, from log-log slope of $F(T)$

Symbol T serves double duty as the paper's canonical service floor and as the window-length argument of the Fano factor $F(T)$; context disambiguates. The per-window packet count and the packet-Hawkes branching ratio would both naturally be written n_{pkt} ; the paper reserves n_{pkt} for the branching ratio and writes $n_{\text{pkt}}^{\#}$ for the count.

Part I

Analytical framework

2 Analytical framework

A baseline proposition, a theorem, and a corollary anchor the paper’s simulation. Proposition 1 is the isolated-burst identity: an N -stage deterministic-service tandem clears a large burst N times faster than one server. Theorem 1 is the general statement: the tandem’s end-to-end wait equals the wait at a single server with the largest stage’s service plus $(N - 1)h$, exactly and on every sample path, and that single-server wait is at most $1/N$ of the wait at service T . Both are elementary and arrival-law independent; the second rests on a reduction theorem of Friedman [11] and on the monotonicity of the Lindley recursion. Corollary 2 is the Poisson null: at the utilisation of a CME packet feed the single-stage wait is zero at every reported quantile, so splitting costs exactly $(N - 1)h$. The arrival law enters only through the size of the single-stage wait that the $1/N$ is applied to, and the Hawkes–Oakes cluster representation explains why that wait is large on a real feed. The two rounds of literature search summarised in Section 1.4 found no published statement of the architecture recommendation flipping with the arrival law; the results below are the paper’s formal content, and Part II measures the slack in the bound on real arrivals.

2.1 Burst-limit throughput theorem

Proposition 1 (Burst-limit N -factor speedup). *Consider an isolated cluster of k arrivals whose j -th member arrives no later than $(j - 1)T/N$ after the first, so that no server idles during the burst (an instantaneous batch is the canonical case). Let $\tau_k^{(1)}$ denote the exit time of the last arrival under a single-stage deterministic-service queue with service T , and let $\tau_k^{(N)}$ denote the exit time under an N -stage tandem with per-stage deterministic service T/N and per-stage hop delay h . Then*

$$\tau_k^{(1)} = kT, \quad \tau_k^{(N)} = T + (k - 1) \frac{T}{N} + (N - 1)h, \quad (1)$$

and in particular

$$\lim_{k \rightarrow \infty} \frac{\tau_k^{(1)}}{\tau_k^{(N)}} = N. \quad (2)$$

Remark. *Proposition 1 is the classical pipelined-tandem throughput result [8, 9]; we state it without proof and use it only to fix the baseline against which the arrival-dependent tail reduction of Theorem 1 is measured. What is paper-specific is the numerical calibration of T and h on a production HFT actor framework and the observation that under real CME MDP3 arrivals the ratio $\tau_k^{(1)}/\tau_k^{(N)}$ is what governs the observed tail-latency compression.*

Reading Equation (1). In the paper’s HFT setting we take $T = 10\mu\text{s}$ as a typical HFT task on a single-thread event loop, and k is the size of a Hawkes-cluster burst that hits the pipeline back-to-back. The single-stage formula $\tau_k^{(1)} = kT$ says that in an event-loop design the last message of a k -burst waits behind $k - 1$ predecessors, each of which spends the full T on the one server; a 10-packet cluster on a $10\mu\text{s}$ server therefore has a last-message latency of $100\mu\text{s}$. The tandem formula $\tau_k^{(N)} = T + (k - 1)(T/N) + (N - 1)h$ decomposes the last-message latency into three physical pieces: (i) T , the total service work every message must traverse regardless of N ; (ii) $(k - 1)(T/N)$,

the pipelined wait behind $k - 1$ predecessors at throughput N/T rather than $1/T$; and (iii) $(N - 1)h$, the one-time hop-cost tax paid to traverse $N - 1$ mailbox boundaries on the way through. The single message ($k = 1$) pays only (i) and (iii), never (ii). The whole point of the paper is that under Hawkes arrivals term (ii) is what blows up on a single stage and shrinks by a factor of N under a tandem, while term (iii) is only a modest additive tax when the framework's h is small.

Reading Equation (2). The ratio $\tau_k^{(1)}/\tau_k^{(N)}$ tends to N as the cluster size k grows, meaning that in the burst-saturation limit a tandem of N actors clears the burst N times faster than an event loop. This is the classical pipeline throughput speedup restated for HFT: a 4-actor pipeline drains a large burst in one-quarter of the wall-clock time that a single-thread event loop would take, and a 12-actor pipeline in one-twelfth. It is worth emphasising that this speedup is achieved with the same total service work per message: the pipeline does not reduce how much decode/book/strategy work happens per event, it only overlaps the work across cores so the queue behind a big cluster clears in parallel. The arrival law does not enter Equation (2); what the arrival law controls is *how often* the burst regime is entered, and Theorem 1 shows the same factor governs every message on every sample path.

On what T is, and on the natural stage boundaries. The paper adopts $T = 10\mu\text{s}$ as a ballpark typical HFT single-thread event loop. Any real HFT pipeline of that duration is already a sequential chain of internally atomic sub-tasks (SBE decode, order-book application, strategy touch, order construction on CME MDP3, per Section 7), so the N stages of the tandem are placed at the existing boundaries between those sub-tasks; the paper is not subdividing any single sub-task, and in particular is not subdividing the SBE decode, which is a linear pass over the wire bytes of a single message and cannot be split. Ten microseconds is a round-number ballpark chosen for exposition; none of the paper's qualitative results depend on it. The relevant knob for the design equation is the ratio T/h : doubling T at fixed h doubles the cluster-wait terms of Equation (1) and leaves the hop tax unchanged, which is why the floor below which splitting cannot pay (Section 4.3) is stated in units of h . The paper writes “per-stage service T/N ” as a simplifying modelling convention that assumes equal-sized stages; in reality the per-stage services s_i sum to T but are unequal, and the achievable pipeline throughput is $1/\max_i s_i$ rather than N/T . Theorem 1 handles the unequal case exactly: the factor becomes $s_{\max}/T$ in place of $1/N$, the tail still contracts under Hawkes, and the median still grows linearly at slope h per hop.

This is the mechanism. It applies to any deterministic-service tandem regardless of the arrival law; the arrival law determines *how often* large- k clusters occur, and the next subsection makes the per-burst factor a pathwise bound.

2.2 Exact reduction of the tandem to one server, and the scaling bound

Proposition 1 treats one isolated burst. The result that carries the paper's design claim is that the same N -factor holds pathwise for every message under every arrival sequence, at the level of the end-to-end wait. Fix an arrival sequence $a_1 \leq a_2 \leq \dots$ (any point process, clustered or otherwise, with batches allowed as coincident times) and write $\delta_j = a_{j+1} - a_j \geq 0$. For a single FIFO server with deterministic service s fed by this sequence, the waiting time of message j obeys the Lindley recursion [12]

$$w_1(s) = 0, \quad w_{j+1}(s) = (w_j(s) + s - \delta_j)^+. \quad (3)$$

Define the end-to-end wait of message j in the N -stage tandem as $W_j^{(N)} = (\text{exit time}) - a_j - T$, i.e. latency in excess of the total service work; under $N = 1$ this is $W_j^{(1)} = w_j(T)$.

Theorem 1 (Reduction and scaling bound). *Consider an N -stage tandem of FIFO servers with deterministic per-stage services $s_1, \dots, s_N$ summing to T , deterministic hop delay h between consecutive stages, unbounded buffers, and an arbitrary arrival sequence. Let $s_{\max} = \max_i s_i$. Then for every message j and every sample path,*

$$W_j^{(N)} = w_j(s_{\max}) + (N-1)h, \quad (4)$$

$$w_j(s') \leq \frac{s'}{s} w_j(s) \quad \text{for all } 0 < s' \leq s, \quad (5)$$

and consequently

$$W_j^{(N)} \leq \frac{s_{\max}}{T} W_j^{(1)} + (N-1)h = \frac{W_j^{(1)}}{N} + (N-1)h \quad \text{under equal stages } s_i = T/N. \quad (6)$$

Since Equation (6) holds pathwise, it holds for every quantile of the stationary distribution whenever one exists: $W_q^{(N)} \leq W_q^{(1)}/N + (N-1)h$ for all q . The same inequality holds for the empirical (sample) quantiles computed over any finite trace, directly from the pathwise bound and without invoking stationarity; the sample-quantile form is what Part II measures, and it should be read as a statement about the observed windows rather than about a stationary-distribution quantile or a finite-sample maximum.

Proof. Equation (4), equal stages. Departures from a FIFO server with deterministic service s are spaced at least s apart, because each departure requires s of uninterrupted service after the previous one. A FIFO server with deterministic service s whose arrivals are spaced at least s apart never queues: by induction, message j departs at its arrival time plus s , which is at or before the arrival of message $j+1$. Stage 1 is a single server with service T/N fed by the original sequence, so message j leaves it at $a_j + w_j(T/N) + T/N$. The arrival sequence at stage 2 is the stage-1 departure sequence shifted by h , spaced at least T/N apart, so stage 2 adds exactly $h + T/N$ and its departures are again spaced at least T/N apart; the same holds at each subsequent stage. Summing, the exit time is $a_j + w_j(T/N) + N \cdot T/N + (N-1)h$, and subtracting $a_j + T$ gives Equation (4) with $s_{\max} = T/N$.

Equation (4), unequal stages. By the order-invariance theorem of Friedman [11], for a tandem of FIFO stages with constant service times, unlimited queues, and any arrival sequence, each customer's time in system is independent of the order of the stages. Deterministic hop delays preserve this: a hop is a constant shift of one stage's departure sequence into the next stage's arrival sequence, and constant shifts commute with the Lindley recursion, so reordering stages together with their hops leaves every exit time unchanged. Reorder so that the stage with service $s_{\max}$ comes first. Its departures are spaced at least $s_{\max}$ apart, and every subsequent stage has service $s_i \leq s_{\max}$, so by the no-queueing argument above each of them adds exactly $s_i + h$ to every message. The exit time is $a_j + w_j(s_{\max}) + \sum_i s_i + (N-1)h$, and subtracting $a_j + T$ gives Equation (4).

Equation (5). By induction on j with $w_1(s') = 0 = (s'/s)w_1(s)$. The map $x \mapsto (x + s' - \delta_j)^+$ is nondecreasing, so

$$w_{j+1}(s') = (w_j(s') + s' - \delta_j)^+ \leq \left(\frac{s'}{s}w_j(s) + s' - \delta_j\right)^+ \leq \left(\frac{s'}{s}w_j(s) + s' - \frac{s'}{s}\delta_j\right)^+ = \frac{s'}{s}w_{j+1}(s),$$

where the second inequality uses $\delta_j \geq (s'/s)\delta_j$, which holds because $\delta_j \geq 0$ and $s' \leq s$. Combining Equations (4) and (5) with $s' = s_{\max}$, $s = T$ gives Equation (6). $\square$

Remark (What the theorem says and what it does not). Equation (4) says all queueing in an equal-stage deterministic tandem happens at stage 1; stages $2, \dots, N$ contribute exactly $T/N + h$

each to every message. Proposition 1 is the isolated-burst special case, where Equation (6) holds with equality. Equation (6) is a bound rather than an identity because the tandem runs its bottleneck stage at utilisation ρ/N while the single stage runs at ρ : the induction step in the proof of Equation (5) gives away $(1 - 1/N)\delta_j$ at every message j that the single server is still serving when message $j + 1$ arrives, so the tandem clears each inter-arrival gap N times more effectively than the scaled bound credits it for. The paper’s empirical γ (Section 4.3), defined by $W_q^{(N)} - (N - 1)h \approx N^{-\gamma} W_q^{(1)}$, therefore satisfies $\gamma \geq 1$ in the model, and the excess $\gamma - 1$ measures that slack. Nothing in the theorem depends on the arrival law; what the arrival law determines is the size of $W_q^{(1)}$ that the factor $1/N$ is applied to, which is the subject of the next remark. The hop cost h enters every quantile as exactly $(N - 1)h$, in the median and in the tail alike.

Remark (Unequal stages: only the largest stage matters). Real cut points do not give equal stages, and Equation (4) says what that costs: the end-to-end wait of any deterministic tandem equals the wait at a single server with service $s_{\max}$, plus h per hop. Three consequences follow, none of which the equal-stage form makes visible.

1. The tail factor is $s_{\max}/T$, and the stage count enters only through the hop tax. Splitting a $10\ \mu\text{s}$ task as $7 + 3$ compresses the tail excess to at most 70% of single-stage, against 50% for $5 + 5$, and both pay one h .
2. The order of the stages is irrelevant to end-to-end latency: $7 + 3$ and $3 + 7$ give identical latencies on every sample path. Order determines only where the queue physically sits (below).
3. A cut that does not lower $s_{\max}$ buys nothing. Refining $7 + 3$ into $7 + 2 + 1$ adds a hop and leaves every quantile of the wait unchanged, so end-to-end latency rises by exactly h . Conversely, sub-tasks can be merged onto one thread (via `fast_send`) without any tail cost as long as the merged service does not exceed $s_{\max}$. The design problem is therefore to choose the partition with the smallest $s_{\max}$ reachable at the natural cut points, and then the fewest hops that realise it.

Appendix C checks all three numerically.

Remark (Where the burst goes: the bottleneck absorbs the clustering). The proof of Equation (4) rests on one fact: departures from a deterministic server with service s are spaced at least s apart. Applied to a Hawkes input, that fact says what the bottleneck stage does to the clustering. Upstream of the bottleneck the stream is the raw feed, with intra-cluster inter-arrival times far below $s_{\max}$ and instantaneous intensity far above $1/s_{\max}$. Downstream of the bottleneck, every cluster still leaves as the same group of K messages, so the count-level branching structure is unchanged (the Borel cluster sizes, and the Fano factor at time scales beyond $Ks_{\max}$, are exactly those of the input), but the intra-cluster spacing has been regularised to $s_{\max}$ and the instantaneous intensity is capped at $1/s_{\max}$. The clustering has not been removed; it has been made harmless to any consumer at least as fast as the bottleneck, which is why stages with service $s_i \leq s_{\max}$ never queue. If the bottleneck is stage 1, all of the waiting in the pipeline sits in its mailbox and every downstream mailbox holds at most one message; if the bottleneck is stage $b > 1$, the stages before it queue partially (the raw burst is still bursty relative to their smaller services) and stage b absorbs the remainder, with the same total by Equation (4).

The practical consequences for the splitting recommendation are these. First, put the largest stage first, at ingress: end-to-end latency is unchanged by the order, but this confines all queueing to one mailbox, keeps the downstream mailboxes at depth one, and places the burst-absorbing queue where its receiver is continuously busy during a cluster (the regime in which the load-dependent hop

cost of Section 4.5 is smallest). Second, any burst-aware logic, such as load shedding, coalescing of superseded book updates, or a cancel-on-burst risk rule, can only observe the raw burst upstream of the bottleneck; downstream it sees a rate- $1/s_{\max}$ stream and cannot tell a cluster from steady flow except by counting. Third, a packet-Hawkes fit run on the bottleneck's departure stream would report a lower intensity peak and a longer effective kernel while recovering the same branching ratio in counts; operators instrumenting a pipeline should fit on the ingress stream. Fourth, deterministic hop delay is what keeps downstream stages queue-free; hop jitter of amplitude ϵ lets a downstream stage queue by at most ϵ per message, so the reduction holds up to that jitter in practice.

Fact (Hawkes–Oakes 1974 cluster representation [2]). *A stationary Hawkes process with background rate $\mu > 0$ and branching ratio $n \in [0, 1)$ is stochastically equivalent to a Poisson-cluster process: a rate- μ Poisson process of immigrant arrivals, each of which independently spawns an offspring cluster whose total size K (including the immigrant) follows the Borel distribution*

$$\Pr(K = k) = \frac{(nk)^{k-1} e^{-nk}}{k!}, \quad k = 1, 2, 3, \dots \quad (7)$$

with mean $E[K] = 1/(1 - n)$ and variance $\text{Var}[K] = n/(1 - n)^3$ [3].

Remark (Where Hawkes enters: the size of the single-stage wait). *Under Poisson arrivals at the utilisation of a CME packet feed ($\rho = \bar{\lambda}_{\text{pkt}}T \ll 1$), the single-stage wait is zero for almost every message, so $W_q^{(1)} = 0$ at every quantile the paper reports and Equation (6) has nothing to compress; that is Corollary 2. Under Hawkes arrivals the cluster representation puts K near-coincident messages behind one immigrant, and in the instantaneous-batch picture the last of them waits $(K - 1)T$ on a single stage. Stirling's formula applied to Equation (7) gives, for large k ,*

$$\Pr(K = k) \approx \frac{1}{n\sqrt{2\pi}} k^{-3/2} (ne^{1-n})^k, \quad (8)$$

and $ne^{1-n} < 1$ for every $n < 1$, with $-\log(ne^{1-n}) \approx (1 - n)^2/2$ near criticality. So the Borel batch size is light-tailed at every fixed $n < 1$: its survival function follows the critical $k^{-1/2}$ power law only in the window $k \ll k^* = (1 - n)^{-2}$ and decays geometrically beyond it (Appendix A). The consequences for the paper are, first, that the largest clusters, and hence the single-stage cluster wait they produce, have a characteristic scale of order $Tk^* = T(1 - n)^{-2}$, which at the corpus-median $n \approx 0.8$ (Section 3.2) is $\approx 25T$ and at $n = 0.9$ would be $100T$, one to two orders of magnitude above the service floor and the hop cost; second, that heavy-tail results built on subexponential batch sizes, such as the single-big-batch principle for $M^{[K]}/G/1$ [4] and the monotone-separable tail asymptotics of Baccelli et al. [14], do not apply to this batch distribution at any fixed $n < 1$, so no closed-form power-law exponent for the tail probability should be expected; and third, that none of this is needed for the design claim, because Theorem 1 applies the factor $1/N$ to whatever $W_q^{(1)}$ the feed produces. The size of $W_q^{(1)}$ on real feeds is a measurement, and Part II reports it.

Corollary 2 (Poisson null). *Under homogeneous Poisson arrivals at rate $\bar{\lambda}$ with $\rho = \bar{\lambda}T < 1 - q$, the q -quantile of the single-stage wait is $W_q^{(1)} = 0$, because $\Pr(W^{(1)} > 0) = \rho$ for $M/D/1$ by PASTA. By Theorem 1, $W_q^{(N)} \leq (N - 1)h$, and by Equation (4) $W_q^{(N)} \geq (N - 1)h$, so*

$$W_q^{(N)} = (N - 1)h \quad \text{for all } N \text{ and all } q \leq 1 - \rho. \quad (9)$$

At the utilisations of a CME packet feed (corpus-median 452 packets per second, so $\rho = 0.0036$ at $T = 8 \mu\text{s}$) this covers p99 at every $T \leq 16 \mu\text{s}$ in the paper's sweep: end-to-end latency is $T + (N - 1)h$

at p_{50} and at p_{99} alike, and splitting costs $(N - 1)h$ with nothing to compress. Where ρ/N does exceed $1 - q$, Equation (6) still holds and the tandem compresses the small Poisson wait too; the sweep contains exactly one such p_{99} cell ($T = 32 \mu\text{s}$, $N = 1$, $\rho = 0.0145$) and several $p_{99.9}$ cells, and Section 4.1 shows the wait appearing in precisely those.

2.3 What the theory says, in plain terms

This subsection restates Theorems 1 and 1 and Corollary 2 without the formalism, for readers who build pipelines rather than analyse queues. The formal statements above remain the reference.

Start from where a message’s time goes. A message’s end-to-end latency is the sum of three things: the work done on it (T in total, however many stages that work is spread over), the hop tax (h for every mailbox boundary it crosses, so $(N - 1)h$ in an N -stage pipeline), and the time it spends sitting in queues waiting for a busy actor. The paper calls that third piece the wait, W . The work is fixed by the code and the hop tax is fixed by the framework; the wait is the only piece the architecture can change, and on a bursty feed it is the piece that dominates the tail.

Proposition 1: one burst. Suppose k messages land at once. On a single thread the last of them waits for the $k - 1$ ahead of it, each costing T : it waits $(k - 1)T$. On an N -stage pipeline the first stage takes only T/N per message, so the last message waits $(k - 1)T/N$ at stage 1 and then flows through the remaining stages without waiting again, because each of them is fed at exactly the rate it can serve. The pipeline clears a burst N times faster for the same total work, because N cores are working on N different messages at the same instant. That is what pipelining means, and it is the whole mechanism.

Theorem 1: every message, whatever the traffic. Two facts, each of which a programmer can check by hand.

1. In a pipeline where each stage does a fixed amount of work per message, only the slowest stage ever builds a queue. Every other stage receives messages no faster than it can process them, because the slowest stage upstream (or, if the slowest stage is downstream, the stages before it) meters them out. So the whole pipeline behaves exactly like one thread whose per-message work is the slowest stage’s work, $s_{\max}$, plus the hop tax. With equal stages, $s_{\max} = T/N$.
2. Take any recording of arrival times, real or synthetic, and replay it through two single-threaded servers: one doing T of work per message and one doing T/N . The second server’s queue wait for message number j is never more than $1/N$ of the first server’s wait for the same message. The reason is that in the faster server every backlog is N times smaller and every idle gap between arrivals clears at least as much of it.

Put together: for every message in every replay, the pipeline’s wait is at most the single-thread wait divided by N , plus $(N - 1)h$. Because that holds message by message, it holds for whichever summary statistic the operator cares about: median, p_{99} , $p_{99.9}$, or the worst case. And the bound is usually beaten. It is met exactly on an isolated burst that arrives all at once, and beaten whenever arrivals are spread out, because the pipeline’s first stage is idle more often than the single thread and uses each idle gap to drain more of its backlog. The paper writes the measured compression as $N^{-\gamma}$; the theorem guarantees $\gamma \geq 1$, and Section 4 measures γ between 1.0 (at $T = 2 \mu\text{s}$, where the bound is tight) and 1.7 (at $T = 32 \mu\text{s}$).

Where the traffic model comes in. Nowhere in the two theorems. The arrival pattern enters only through the size of the single-thread wait that the factor $1/N$ is applied to. Under Poisson traffic at the load of a CME packet feed, the single thread is idle when almost every message arrives, so its wait is zero at every quantile the paper reports; there is nothing to divide by N , and the pipeline is simply $(N - 1)h$ slower (Corollary 2). Under the real feed, packets arrive in self-exciting clusters, the single thread’s p_{99} is 2.7 to 8 times T in the paper’s data for tasks of 4 to $32\mu\text{s}$, and the pipeline divides the excess by N or better. The Hawkes cluster representation says why the clusters are large: a cluster’s size follows the Borel distribution, whose typical large cluster has of order $(1 - n)^{-2}$ packets, about 25 at the corpus’s $n \approx 0.8$.

The rules a designer can take away.

- The tail improvement is set by the slowest stage, $s_{\max}/T$, and by nothing else about the decomposition. Balanced stages are the best case for a given number of hops.
- A cut that does not shrink the slowest stage costs one h at every quantile and improves nothing. Sub-tasks can be merged onto one thread until their combined work reaches $s_{\max}$ at no tail cost.
- The order of the stages does not change any message’s latency. It changes only where the queue physically sits; putting the slowest stage first keeps all queueing in one mailbox.
- Every hop costs exactly h on every message, median and tail alike; there is no regime where the hop tax is hidden.
- Whether splitting is worth it comes down to one comparison: the single-thread wait at the quantile you care about, times $(1 - s_{\max}/T)$, against h per hop added. On a Poisson-like feed the left side is zero; on a clustered feed it is tens of microseconds.

The guarantees above are exact inside a specific model, and a designer should know what that model leaves out. The rest of this section lists its assumptions and the questions it leaves to measurement.

Fixed work per message. Each stage is assumed to spend the same amount of time on every message. That is what makes only the slowest stage queue: a stage that sometimes takes longer than its upstream neighbour would occasionally fall behind and build its own queue, and then the pipeline’s wait would exceed the single-server wait at $s_{\max}$. The paper’s simulator uses constant service, so the theorems describe the simulator exactly; real actors have data-dependent work (a book update touching a deep level costs more than one touching the top), and for them the theorems are a model to be checked by measurement, which is the paper’s operational recommendation in any case (Section 7).

No drops, no bounds, in-order. Mailboxes are unbounded, messages are never discarded or coalesced, and each stage processes in arrival order. A framework that drops or merges messages under load changes the arithmetic (in the pipeline’s favour, at the cost of lost messages); a priority queue changes which message pays the wait but not the total.

Fixed hop delay. The hop is modelled as a constant h . A jittery hop lets a downstream stage queue by at most the jitter amplitude per message, so the reduction holds up to that amplitude. The paper’s calibration uses the low-load, wake-from-sleep value of h , which Section 4.5 argues is an upper bound inside a burst.

Proposition 1 needs a dense burst. The one-burst identity assumes the k messages arrive faster than the first stage can drain them, i.e. faster than one per T/N . For a burst that trickles in, the ratio in Equation (2) is $kN/(k + N - 1 + N(N - 1)h/T)$, which climbs toward N as the burst grows. Theorem 1 does not need this assumption; it covers every arrival pattern.

What the theory does not predict. It does not say how large the single-thread wait is on a given feed. That number depends on the feed’s clustering and utilisation and, as the remark after the Borel fact explains, has no clean closed form at a fixed branching ratio: the Borel cluster size is light-tailed for every $n < 1$, so the heavy-tail formulas that would give a power-law exponent do not apply. The paper measures the single-thread wait on real CME arrivals instead (Part II). Nor does the theory predict how far below the $1/N$ bound the pipeline lands; that slack is also measured. What the theory does guarantee is that the pipeline can never do worse than $1/N$ of the single-thread wait plus the hop tax, on any feed whatsoever, which is the statement a designer needs in order to trust the trade before measuring it.

Part II

Empirical experiment

3 Experiment setup

This section describes the empirical experiment that tests the analytical predictions of Section 2: the data corpus, the confirmation that packet arrivals are Hawkes-clustered (the reason the single-stage wait that Theorem 1 divides by N is large at all), the tandem-Lindley simulator, and the sweep design that produces the numbers reported in Section 4. Figure 1 shows the end-to-end data flow of a single 30-minute window; Algorithm 1 states the full per-window procedure in pseudocode.

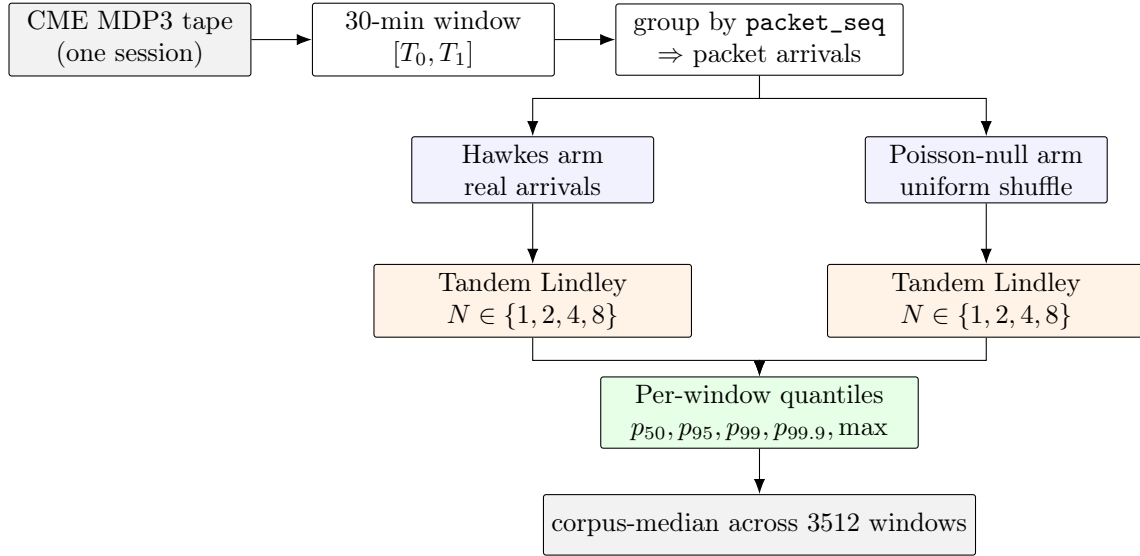


Figure 1. Per-window data flow of the empirical experiment. A single session tape is chopped into 30-minute RTH windows. Each window’s messages are grouped by `packet_seq` into a real packet-arrival stream (Hawkes arm). A rate-matched Poisson-null arm is generated by uniformly shuffling the same arrival count over the same window. Both arms are run through the N -stage tandem Lindley simulator at $N \in \{1, 2, 4, 8\}$. Per-window quantiles are recorded and aggregated by corpus median across all 3512 windows in the 276-session corpus.

Algorithm 1 Per-window procedure of the empirical experiment.

```
1: input: session tape  $S$ , service  $T$ , hop  $h$ , stage counts  $\mathcal{N}$ 
2: for each 30-min RTH window  $W$  in  $S$  with  $[T_0, T_1]$  do
3:    $\{m_i\} \leftarrow$  messages in  $W$ 
4:    $\{(\text{seq}, t)\} \leftarrow \{(\text{packet\_seq}(m_i), \text{transactTime}(m_i))\}$ 
5:    $\text{arr}_H \leftarrow \text{sort}(\{\min_i t_i : \text{seq}_i = s\}_s)$   $\triangleright$  Hawkes arm (real arrivals)
6:    $n_{\text{pkt}}^\# \leftarrow |\text{arr}_H|$ 
7:    $\text{arr}_P \leftarrow \text{sort}(\text{Uniform}(\min \text{arr}_H, \max \text{arr}_H)^{n_{\text{pkt}}^\#})$   $\triangleright$  Poisson-null arm (seeded per window)
8:   for regime  $\in \{H, P\}$ ,  $N \in \mathcal{N}$  do
9:      $\text{lat} \leftarrow \text{TANDEMLINDLEY}(\text{arr}_{\text{regime}}, N, T, h)$ 
10:    record  $(S, W, \text{regime}, N, \{p_{50}, p_{95}, p_{99}, p_{99.9}, \max\}(\text{lat}))$ 
11:   end for
12: end for
13: return corpus-median of each recorded quantile across all  $(S, W)$  pairs
14: function TANDEMLINDLEY( $\text{arr}, N, T, h$ )
15:    $\text{dep} \leftarrow \text{arr}$ 
16:   for stage  $k = 1, \dots, N$  do
17:     if  $k > 1$  then  $\text{dep} \leftarrow \text{dep} + h$ 
18:   end if
19:    $\text{dep} \leftarrow \text{dep} + \text{LINDLEY}(\text{dep}, T/N)$ 
20: end for
21:   return  $\text{dep} - \text{arr}$ 
22: end function
```

3.1 Data corpus

The corpus is 281 sessions of CME MDP3 channel 318 (NQ) front-month pcap-derived tapes covering 2025-01 through 2026-02, RTH only, holidays and half-days excluded. Each MBO record carries a nanosecond `transactTime` (the CME matching-engine event time in the MDP3 message body) and a `packet_seq` (the UDP packet sequence number). Messages are grouped into packets by `packet_seq`; a packet’s arrival time is the minimum `transactTime` of its constituent events. Each session tape is chopped into non-overlapping 30-minute RTH windows; for each window we compute the packet-arrival stream and the per-window quantities listed in Section 1.6. The corpus yields 3512 per-window records from the 276 sessions that contain at least one complete RTH window, roughly 13 windows per session; the remaining five of the 281 tapes are holiday or half-day sessions with none.

3.2 Packet arrivals are Hawkes-clustered (empirical prerequisite)

The paper’s simulator input is the packet-arrival stream, so the first empirical question is whether that stream is clustered or Poisson. Three model-free and one model-based diagnostic on the 276-session, 3512-window NQ front-month corpus all reject Poisson decisively. The fraction of interarrival gaps shorter than one-tenth of the mean gap is 40–85% per 30-minute window, versus the Poisson benchmark $1 - e^{-0.1} \approx 9.52\%$ that would obtain for any exponential-gap process regardless of rate. The Fano factor $F(T) = \text{Var}[C_T]/\text{E}[C_T]$ of packet counts in windows of length T rises from $F(0.1\text{ s}) \approx 2$ to $F(60\text{ s}) \approx 40$ across the corpus median, versus $F(T) \equiv 1$ for Poisson at every T ; the log-log fit gives a Hurst exponent $H \approx 0.65\text{--}0.75$, versus $H = 0.5$ for Poisson. Fitting the

standard exponential-kernel Hawkes process $\lambda(t) = \mu + \sum_{t_i < t} \alpha e^{-\beta(t-t_i)}$ to each window by MLE (Ogata log-likelihood with the standard recursive intensity, closed-form compensator, Nelder–Mead over (μ, α, β)) gives a branching ratio $n = \alpha/\beta$ with a corpus median of 0.794 and a 5th–95th percentile range of 0.757–0.827 across the 3512 windows (extremes 0.567 and 1.000, the latter a single window pinned at the fit boundary), with every fit converged; the same optimum is recovered by a reparameterised L-BFGS-B fit to four decimals on spot-checked windows. This is the same near-critical regime that Filimonov and Sornette [24] and Hardiman et al. [25] document on ES, at a somewhat lower level than their $n \approx 0.9$; they fit trade arrivals rather than packet arrivals, and a packet aggregates several book events, which is expected to lower the fitted self-excitation. Fisher–Yates shuffle of the interarrival gap sequence within each window destroys the clustering entirely — shuffled Fano at $T = 5$ s collapses to ≈ 1 — confirming that the excess variance lives in the ordering, not in the marginal gap distribution (i.e., Hawkes rather than a heavy-tailed renewal process). The paper’s downstream simulator is therefore driven by an arrival process that is empirically far from Poisson, and any tail-latency finding based on a matched-rate Poisson null represents a hard counterfactual, not a straw man. We use the Hawkes fit only as a parsimonious description of the observed clustering; the tandem reduction of Section 2 is arrival-law agnostic and does not depend on the Hawkes assumption — any arrival process with the same clustering drives the same single-server queue that the bound compresses, which is precisely why the model-free diagnostics above, not the Hawkes fit, carry the empirical premise.

3.3 Simulator

Packet arrivals $t_1 < t_2 < \dots$ enter stage 1. Each stage k is a single-server FIFO queue with deterministic service T/N (own actor, own thread), and departures from stage k are delayed by h before arriving at stage $k + 1$. Standard Lindley recursion per stage:

$$W_{i+1}^{(k)} = \max\left(0, W_i^{(k)} + \frac{T}{N} - (a_{i+1}^{(k)} - a_i^{(k)})\right), \quad (10)$$

with $a_i^{(k+1)} = a_i^{(k)} + W_i^{(k)} + T/N + h$.

3.4 Pipelining is what makes multi-stage work

The tandem is a *pipelined* design: each stage’s actor runs on its own thread, so at the same wall-clock instant stage 2 can be processing message $k - 1$ while stage 1 is processing message k . The simulator captures this implicitly because Lindley is applied separately per stage and each stage tracks its own “when did the previous message leave” timeline — the two servers’ busy intervals overlap on the wall-clock axis.

Worked example. Three packets arriving at $t = 0, 1, 2 \mu\text{s}$; single-stage total service $T = 8 \mu\text{s}$ (a service floor in the sweep of Table 1); two-stage tandem with per-stage service $T/2 = 4 \mu\text{s}$ and hop $h = 1.7 \mu\text{s}$:

	msg 1	msg 2	msg 3
single-stage arrival	0	1	2
single-stage latency (μs)	8.0	15.0	22.0
tandem arrive stage 1	0.0	1.0	2.0
tandem wait stage 1	0.0	3.0	6.0
tandem leave stage 1	4.0	8.0	12.0
tandem arrive stage 2 (+ h)	5.7	9.7	13.7
tandem wait stage 2	0.0	0.0	0.0
tandem leave stage 2	9.7	13.7	17.7
tandem latency (μs)	9.7	12.7	15.7

At wall-clock $t \approx 7$, stage 1’s actor is busy on message 2 while stage 2’s actor is busy on message 1 — two actors doing useful work at the same time. Message 1 pays a $1.7\mu\text{s}$ hop cost (latency $8.0 \rightarrow 9.7$); message 3, at the burst tail, drops from 22.0 to $15.7\mu\text{s}$. Median hurts, tail wins — same message count, throughput $2/T$ rather than $1/T$. This is exactly the mechanism of Proposition 1.

Pipelining is a necessary condition, not a nice-to-have. The tail-latency improvement in Proposition 1 and Theorem 1 comes entirely from concurrent execution across the N stages: at any wall-clock instant, several different messages are simultaneously in flight at different stages of the pipeline, and it is that overlap that raises system throughput from $1/T$ to $\min_i 1/s_i$ (equal to N/T under equal-sized stages). Take away the concurrency — for instance by chaining N synchronous handoffs on a single thread, where the sender blocks until each downstream stage finishes — and the entire benefit disappears. In that case each message serially traverses all N stages before the next message can start; total service work per message is still T , so throughput remains $1/T$, and the pipeline adds $(N - 1)h$ of hop cost with no compensating tail win. Under Hawkes bursts a synchronous N -stage chain is strictly worse than a single-stage event loop by exactly $(N - 1)h$ per message. Pipelining — concurrent execution of different messages at different stages on different threads — is therefore not an optimisation on top of the tandem design, it is the design; the paper’s simulator captures this by applying Lindley separately per stage, giving each stage its own busy-idle timeline. The framework-implementation Section 7 discusses which KASPAR-HFT dispatch paths actually deliver this concurrency (the asynchronous BQueue path) and which do not (the synchronous `fast_send` path, which despite its low per-call cost does not support pipelining).

Calibration and cost. The paper reports numerical results at $h = 1.7\mu\text{s}$ — the one-way async mailbox cost obtained by halving the round-trip figure in Table 4 of Maciejewski [33] and consistent with the shipping KASPAR-HFT BQueue path characterised in Maciejewski [32] — swept across $T \in \{2, 4, 8, 16, 32\}\mu\text{s}$ to cover the sub-hop regime through the slow-servicing regime. The simulator is deterministic given the tape and (T, h, N) ; one $O(n)$ pass per stage per window makes its wall-clock cost negligible next to the tape read.

3.5 Sweep design

The design asks whether an N -stage tandem reduces p_{99} latency under real Hawkes packet arrivals and whether the same architecture achieves the same compression under a rate-matched Poisson null. For each 30-minute window we run the tandem Lindley of Section 3.3 at $N \in \{1, 2, 4, 8\}$ under two arrival regimes: the real packet-arrival timestamps from the tape (Hawkes arm) and a uniform-shuffle of the same $n_{\text{pkt}}^\#$ arrivals over the same support (Poisson-null arm). The support is

the observed packet span $[\min \text{arr}_H, \max \text{arr}_H]$ rather than the nominal window $[T_0, T_1]$, so the null is rate-matched to the arrivals it replaces rather than to wall-clock; the two coincide on busy windows and the former is the stricter counterfactual on sparse ones. The null is therefore a *conditional homogeneous-Poisson* process: a homogeneous Poisson process at rate $n_{\text{pkt}}^\# / (\max \text{arr}_H - \min \text{arr}_H)$ conditioned on the observed count over the observed support. Per (window, N , regime) we record $p_{50}, p_{95}, p_{99}, p_{99.9}$, max of end-to-end latency. Aggregation is corpus-median across all windows; the two headline curves per scenario are $p_{99}(N)$ and $p_{50}(N)$ under Hawkes and Poisson overlaid.

The paper’s empirical calibration is $h = 1.7 \mu\text{s}$ (one-way async cross-thread mailbox cost, per Maciejewski [33]’s Table 4 halved for one-way) swept over $T \in \{2, 4, 8, 16, 32\} \mu\text{s}$ so the scenarios span from the sub-hop regime through the slow-servicing regime. Theorem 1 predicts $p_{99}^H(N) - (N - 1)h - T \leq (p_{99}^H(1) - T)/N$ on every window, i.e. a fitted exponent $\gamma \geq 1$ in $\Delta(N) \approx \Delta(1)N^{-\gamma}$; Corollary 2 predicts $p_{99}^P(N) = p_{50}^P(N) = T + (N - 1)h$ exactly; any of (i) a window on which the Hawkes bound is violated, (ii) Poisson $p_{99}(N)$ exceeding $T + (N - 1)h$, or (iii) Hawkes $p_{50}(N)$ growing at slope $\gg h$ per stage falsifies the paper’s model of the simulator.

4 Empirical results

4.1 Tail collapse under Hawkes, and its absence under Poisson

Table 1 reports corpus-median end-to-end latency quantiles for every (T, N) cell of the sweep at $h = 1.7 \mu\text{s}$, on 276 sessions and 3512 thirty-minute windows (five of the 281 session tapes contain no complete RTH window and are dropped). Figure 2 plots the same numbers. The Poisson-null columns are the uniform-shuffle counterfactual of Section 3.5 with the same packet count per window.

T	N	Hawkes (real arrivals)			Poisson null		
		p_{50}	p_{99}	$p_{99.9}$	p_{50}	p_{99}	$p_{99.9}$
2	1	2.00	4.00	7.38	2.00	2.00	2.00
2	2	3.70	4.70	5.52	3.70	3.70	3.70
2	4	7.10	7.60	7.75	7.10	7.10	7.10
2	8	13.90	14.15	14.15	13.90	13.90	13.90
4	1	4.00	10.87	20.00	4.00	4.00	5.78
4	2	5.70	7.70	11.08	5.70	5.70	5.70
4	4	9.10	10.10	10.92	9.10	9.10	9.10
4	8	15.90	16.40	16.55	15.90	15.90	15.90
8	1	8.00	28.20	55.05	8.00	8.00	13.79
8	2	9.70	16.57	25.70	9.70	9.70	11.48
8	4	13.10	15.10	18.48	13.10	13.10	13.10
8	8	19.90	20.90	21.72	19.90	19.90	19.90
16	1	16.00	79.22	157.07	16.00	16.00	29.84
16	2	17.70	37.90	64.75	17.70	17.70	23.49
16	4	21.10	27.97	37.10	21.10	21.10	22.88
16	8	27.90	29.90	33.28	27.90	27.90	27.90
32	1	32.00	254.73	497.15	32.00	42.02	62.02
32	2	33.70	96.92	174.77	33.70	33.70	47.54
32	4	37.10	57.30	84.15	37.10	37.10	42.89
32	8	43.90	50.77	59.90	43.90	43.90	45.68

Table 1. Corpus-median end-to-end latency quantiles in μs at $h = 1.7 \mu\text{s}$, for each service floor T (μs) and stage count N . 276 sessions, 3512 thirty-minute RTH windows of CME MDP3 NQ front-month packet arrivals. The Poisson null is the uniform shuffle of the same arrival count per window.

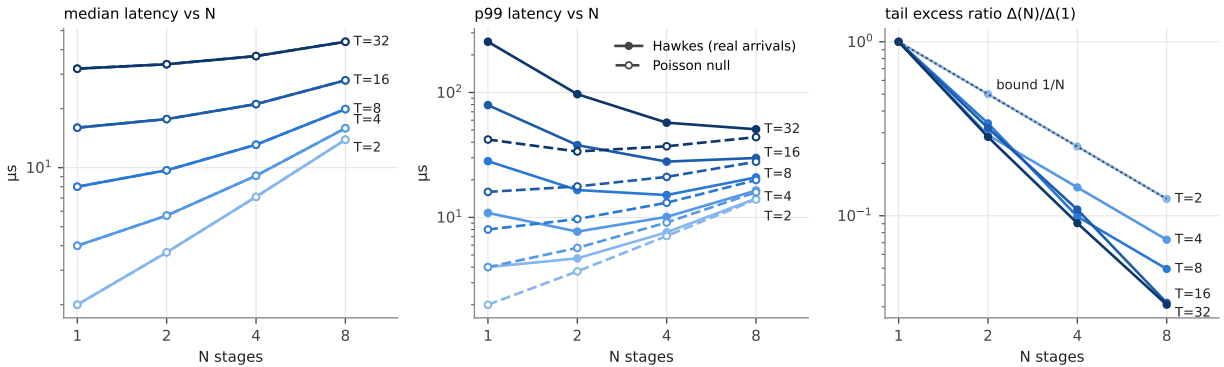


Figure 2. Corpus-median latency versus stage count N for each service floor T (one shade per T , labelled at the right). Left: median. Centre: p_{99} , Hawkes solid and Poisson null dashed. Right: the Hawkes tail excess ratio $\Delta(N)/\Delta(1)$ against the Theorem 1 bound $1/N$ (dotted). The median rises by exactly h per hop in both arms; the Hawkes p_{99} falls with N at every $T \geq 4 \mu\text{s}$ while the Poisson p_{99} tracks the median.

Under the Poisson null the queue is empty when almost every message arrives, so every reported quantile equals the service floor, $p_{50} = p_{99} = T + (N - 1)h$, in every cell with $T \leq 16 \mu\text{s}$. The single exception is $T = 32 \mu\text{s}$, $N = 1$, where Poisson p_{99} is $42.0 \mu\text{s}$ against a floor of 32: at the corpus-median packet rate of 452 packets per second the utilisation there is $\rho = 452 \times 32 \mu\text{s} = 0.0145$,

above the 0.01 at which Corollary 2 says the Poisson wait must first appear at p_{99} , and it does; at $N = 2$ the per-stage utilisation halves to 0.007 and the wait vanishes again. The $p_{99.9}$ column tells the same story one decade down: the Poisson wait is visible at $N = 1$ for every $T \geq 4 \mu\text{s}$ (single-stage $\rho > 10^{-3}$) and at $N = 2$ for $T \geq 8 \mu\text{s}$, and absent wherever the per-stage ρ/N falls below 10^{-3} . The null arm is not inert; it is a queue with a small, predictable wait, and splitting under Poisson loses $(N - 1)h$ at every quantile the operator would look at.

Under Hawkes the story is different at every service floor above the hop cost. Single-stage p_{99} is 2.0, 2.7, 3.5, 5.0, and 8.0 times the service floor at $T = 2, 4, 8, 16, 32 \mu\text{s}$, and single-stage $p_{99.9}$ reaches $15.5T$ at $T = 32$. Splitting into two stages cuts the p_{99} from 10.87 to $7.70 \mu\text{s}$ at $T = 4$, from 28.2 to 16.6 at $T = 8$, from 79.2 to 37.9 at $T = 16$, and from 254.7 to 96.9 at $T = 32$, in each case for a median cost of exactly $1.7 \mu\text{s}$; the two-stage p_{99} is below the single-stage p_{99} on 100% of the 3512 windows at every $T \geq 4 \mu\text{s}$. The stage count that minimises p_{99} rises with T : $N = 2$ at $T = 4$, $N = 4$ at $T = 8$ and 16, $N = 8$ at $T = 32$. At $T = 2 \mu\text{s}$, below the $2h$ floor, splitting loses on p_{99} (4.70 against 4.00) and wins on only 27% of windows; the row is in the sweep to demonstrate that floor, and it does.

4.2 Median cost is linear in N , exactly as $(N - 1)h$

The p_{50} columns of Table 1 match the single-message formula of Proposition 1 in every one of the twenty cells, under both arms:

$$p_{50}(N) = T + (N - 1)h \quad \text{with } h = 1.7 \mu\text{s},$$

for instance 8.00, 9.70, 13.10, 19.90 at $T = 8$. Under Poisson the identity holds because the queue is empty; under Hawkes it holds because the median message arrives in a quiet inter-cluster stretch where the queue is also empty. The hop tax is therefore exactly h per boundary on the median, with no hidden component, as Theorem 1 says it must be at every quantile.

4.3 Design equation

From the corpus:

$$p_{99}^{\text{Hawkes}}(N) = T + (N - 1)h + \Delta(N), \tag{11}$$

$$p_{50}^{\text{Hawkes}}(N) = T + (N - 1)h, \tag{12}$$

where $\Delta(N) \equiv p_{99}^{\text{Hawkes}}(N) - p_{50}^{\text{Hawkes}}(N)$ is the “clustering tail excess”. Table 2 lists it, together with the ratio $\Delta(N)/\Delta(1)$ that Theorem 1 bounds by $1/N$.

T	$\Delta(N)$ (μs)				$\Delta(N)/\Delta(1)$			γ
	$N=1$	$N=2$	$N=4$	$N=8$	$N=2$	$N=4$	$N=8$	
2	2.00	1.00	0.50	0.25	0.500	0.250	0.125	1.00
4	6.87	2.00	1.00	0.50	0.291	0.146	0.073	1.23
8	20.20	6.87	2.00	1.00	0.340	0.099	0.050	1.48
16	63.22	20.20	6.87	2.00	0.320	0.109	0.032	1.65
32	222.73	63.22	20.20	6.87	0.284	0.091	0.031	1.67
bound					0.500	0.250	0.125	≥ 1

Table 2. Corpus-median clustering tail excess $\Delta(N) = p_{99}(N) - p_{50}(N)$ under Hawkes arrivals at $h = 1.7 \mu\text{s}$, its ratio to the single-stage value, and the least-squares exponent γ in $\Delta(N) \approx \Delta(1)N^{-\gamma}$. The last row is the Theorem 1 bound.

Two features of Table 2 are the theorems made visible. First, Δ depends on (T, N) only through the per-stage service T/N : every diagonal of constant T/N carries one number (1.00 at $T/N = 1 \mu\text{s}$, 2.00 at 2, 6.87 at 4, 20.20 at 8), to the reported precision, across up to four different (T, N) pairs. That is the reduction identity Equation (4) on real arrivals: the tandem’s tail is the single-server tail at $s_{\max}$, and the table is really a table of $\Delta_{\text{single}}(s)$ at $s \in \{0.25, 0.5, 1, 2, 4, 8, 16, 32\} \mu\text{s}$. Second, every ratio sits at or below the bound $1/N$, and the pathwise check of Section 5 finds no message on any window above it. At $T = 2 \mu\text{s}$ the bound is met exactly ($\gamma = 1.00$): the p_{99} message there waits behind exactly one predecessor, so $\Delta(1) = T$ and the tandem can do no better than divide that one service by N . As T grows the queues lengthen, idle gaps between clusters drain proportionally more of the tandem’s smaller backlog, and γ rises to 1.67 at $T = 32$; that slack is the utilisation effect described after Theorem 1, and Appendix B reproduces it on synthetic Borel clusters.

The operator’s stage-count optimum is

$$N^* = \arg \min_N \left[(N-1)h + \kappa \Delta(N) \right]$$

for the operator’s tail weight κ (the number of microseconds of median the operator will spend to remove one microsecond of p_{99} excess). Over the simulated grid $N \in \{1, 2, 4, 8\}$ at $h = 1.7 \mu\text{s}$ the breakpoints are:

T (μs)	$N^* = 1$	$N^* = 2$	$N^* = 4$	$N^* = 8$
2	$\kappa < 1.7$	$1.7 \leq \kappa < 6.8$	$6.8 \leq \kappa < 27$	$\kappa \geq 27$
4	$\kappa < 0.35$	$0.35 \leq \kappa < 3.4$	$3.4 \leq \kappa < 14$	$\kappa \geq 14$
8	$\kappa < 0.13$	$0.13 \leq \kappa < 0.70$	$0.70 \leq \kappa < 6.8$	$\kappa \geq 6.8$
16	$\kappa < 0.04$	$0.04 \leq \kappa < 0.26$	$0.26 \leq \kappa < 1.4$	$\kappa \geq 1.4$
32	$\kappa < 0.01$	$0.01 \leq \kappa < 0.08$	$0.08 \leq \kappa < 0.51$	$\kappa \geq 0.51$

An operator who values tail and median equally ($\kappa = 1$) lands on $N^* = 1$ at $T = 2$, 2 at $T = 4$, 4 at $T = 8$, 4 at $T = 16$, and 8 at $T = 32 \mu\text{s}$, which is also the ordering of the raw p_{99} minima in Table 1.

At real cut points the stages are unequal, and by Remark 2.2 the tail excess depends on the partition P of the servicing chain only through its largest stage, $\Delta(P) = \Delta_{\text{single}}(s_{\max}(P))$, i.e. the single-stage tail excess measured at service $s_{\max}$. The optimum over partitions is then

$$P^* = \arg \min_P \left[\text{hops}(P)h + \kappa \Delta_{\text{single}}(s_{\max}(P)) \right],$$

which Table 2 lets the operator evaluate directly, since its diagonals are $\Delta_{\text{single}}(s)$ on a power-of-two grid of s . Two rules of thumb fall out. A cut is worth its hop only if it lowers s_{max} by enough that $\kappa [\Delta_{\text{single}}(s_{\text{max}}) - \Delta_{\text{single}}(s'_{\text{max}})] > h$; and among partitions with the same s_{max} the one with the fewest hops wins outright, so sub-tasks downstream of the bottleneck should be merged onto one thread until their sum reaches s_{max} .

Design principle (only s_{max} matters). For a servicing chain cut at fixed boundaries, the tail-latency question is not “how many threads?” but “what is the smallest achievable largest stage s_{max} ?” By Theorem 1 the clustering tail excess of any partition is $\Delta_{\text{single}}(s_{\text{max}})$, independent of the stage count; a cut that does not lower s_{max} compresses no tail and still costs one hop h . Choose the partition with the smallest reachable s_{max} , then the fewest hops that realise it.

Floor on T : short tasks cannot be usefully split. Equations (11)–(12) carry an obvious lower bound on the tasks the paper’s recommendation applies to. Splitting a task of total service T into N stages spends $(N - 1)h$ at every quantile to compress a tail excess $\Delta(1)$ that is itself a small multiple of T (2 to $7T$ on this corpus, Table 2). When T is of order the hop cost h , one hop already costs as much as the entire tail excess it could remove, and the design equation prescribes $N^* = 1$ mechanically. The sweep shows exactly this: at $T = 2\mu\text{s}$, $\Delta(1) = 2.00\mu\text{s}$ and the best a split can do is remove half of it ($1.00\mu\text{s}$) for a cost of $1.7\mu\text{s}$, so p_{99} rises from 4.00 to 4.70; at $T = 4\mu\text{s}$, $\Delta(1) = 6.87\mu\text{s}$, the split removes 4.87 of it for the same 1.7, and p_{99} falls from 10.87 to 7.70 on every window. On this framework’s shipping BQueue path ($h \approx 1.7\mu\text{s}$) the floor is therefore $T \gtrsim 2h \approx 3.4\mu\text{s}$; $T = 4\mu\text{s}$ sits just above it and already wins. Sub-microsecond tasks, for instance a single SBE integer decode or a small handler that fires from a strategy fast path, belong on `fast_send` rather than on a cross-thread mailbox, regardless of feed statistics.

4.4 Across windows

The corpus medians understate the tail on the busiest windows. Across the 3512 windows the single-stage p_{99}/T has a median of 2.0, 2.7, 3.5, 5.0, 8.0 and a 95th percentile of 2.8, 3.3, 4.4, 6.2, 10.7 at $T = 2, 4, 8, 16, 32\mu\text{s}$, so one window in twenty at $T = 32$ has a single-stage p_{99} above $340\mu\text{s}$. The tandem’s win is not confined to the median window: $p_{99}(2) < p_{99}(1)$ holds on 100% of windows for every $T \geq 4$, and $p_{99}(4) < p_{99}(1)$ on 74.5% of windows at $T = 4$ and 100% at every $T \geq 8$. The per-window fitted branching ratio has median 0.794 and a 5th–95th percentile range of 0.757–0.827 (Section 3.2); the tail excess tracks it in the expected direction, with the high- n windows supplying the upper tail of the p_{99}/T distribution. The intervening p_{95} quantile is recorded per window in the sweep (Section 3.5) and omitted from the tables above only for brevity; it lies between the reported p_{50} and p_{99} in every cell by construction, and under the Poisson null it equals the service floor wherever p_{99} does, since $\rho/N < 1 - 0.99$ implies $\rho/N < 1 - 0.95$. The 3512 windows are treated as evaluation units for the descriptive statistics reported here; they are nested within 276 sessions and are not independent, so any statistical uncertainty on the aggregates would be clustered at the session level rather than treating 3512 as an independent sample size.

4.5 Sensitivity to h

The design equation is monotone in h : at fixed T and fixed corpus $\Delta(N)$, every κ breakpoint in the table above scales linearly with h , so halving h halves every breakpoint and moves N^* up one notch at every T for a fixed κ . At $h = 1.7\mu\text{s}$ the p_{99} -minimising N is 1, 2, 4, 4, 8 across the five service

floors; at a hypothetical sub-microsecond busy-poll mailbox it would be 2, 4, 8, 8, 8 and the natural cut-point ceiling of 4–5 stages would bind before the design equation did. For any other framework the procedure is the same: measure the one-way asynchronous hop cost h on its own dispatch path, put it into Equation (11) with the corpus Δ of Table 2, and read off N^* ; a framework whose h exceeds the tail excess a split could remove gets $N^* = 1$, and the recommendation reverses.

h is load-dependent; $1.7\ \mu\text{s}$ is a conservative upper bound. The $h = 1.7\ \mu\text{s}$ figure is the one-way cost of the shipping KASPAR-HFT BQueue mailbox measured at low load, where the receiver is sleeping on a condition variable and the send path must pay the kernel wake latency to bring it back onto CPU. That wake latency is the dominant term in the $\sim 3.4\ \mu\text{s}$ round-trip of Table 4 in Maciejewski [33], and the general fact that a spinning receiver delivers a message an order of magnitude faster than a wake-from-condvar receiver is standard in the microsecond-latency systems literature: David et al. [60] measure the primitive costs directly on commodity Linux hardware, Belay et al. [61] exploit this in the IX data plane by pinning busy-polling network threads, and Rigtorp’s operational guide [39] formalises the “run one thread per core and busy-wait” rule that flows from it; the fast_send paper’s own grouped-send row (Table 4, $\sim 90\ \text{ns}$) is the same effect on the same framework — when the receiver never actually sleeps, the send path collapses. Under sustained load, and specifically inside a Hawkes cluster where the receiver’s queue is never empty, the receiver never sleeps: successive messages arrive faster than the receiver’s per-message service time, so it stays awake in a tight run-loop and the enqueue path reduces to a mutex acquire, a queue push, and a condition-variable notify against a receiver that is not waiting (a near-no-op). The effective h in that regime is dominated by the mutex + cache-line-transfer cost, typically a few hundred nanoseconds rather than $1.7\ \mu\text{s}$. This is exactly the operating regime in which the paper’s tail-latency win is largest — the win compounds while h shrinks — so the paper’s reported effect underestimates the true benefit whenever the pipeline is inside a burst. Characterising the load-conditioned hop cost $h(\rho)$ is left to future work; a full accounting would strengthen the results reported here.

5 Evaluation: do empirics match the analytical predictions?

This section compares the empirical numbers of Section 4 against the closed-form predictions of Section 2.

Proposition 1 (burst-limit throughput). The single-message identity $p_{50}(N) = T + (N - 1)h$ is predicted exactly by Proposition 1 for any arrival law when the median message finds the queue empty. Table 1 confirms it in all twenty (T, N) cells under both arms at $h = 1.7\ \mu\text{s}$, to the $0.01\ \mu\text{s}$ reporting precision.

Theorem 1 (reduction and scaling bound). The prediction is $\Delta(N) \leq \Delta(1)/N$ on every window, and more strongly $W_j^{(N)} \leq W_j^{(1)}/N + (N - 1)h$ for every individual message. The corpus-median ratios in Table 2 sit at or below $1/N$ in every cell, with equality at $T = 2\ \mu\text{s}$ and slack growing with T (γ from 1.00 to 1.67). The pathwise form was checked directly by the simulator: across all 3512 windows, all five service floors, both arms, and $N \in \{2, 4, 8\}$, no message violates the bound (the check allows N nanoseconds for the rounding of T/N ; the largest excess over that tolerance is 0). The reduction identity Equation (4) is visible in Table 2 without any fitting: Δ is constant along every diagonal of constant T/N , so the tandem’s tail is the single-server tail at $s_{\max}$ on real arrivals exactly as on synthetic ones (Appendices B and C).

Corollary 2 (Poisson null). The prediction is $p_q^P(N) = T + (N - 1)h$ wherever the per-stage utilisation satisfies $\rho/N < 1 - q$, and a visible wait wherever it does not. Table 1 matches both halves: Poisson p_{99} equals the floor in nineteen of twenty cells and exceeds it only at $T = 32 \mu\text{s}$, $N = 1$ ($\rho = 0.0145 > 0.01$), and Poisson $p_{99.9}$ exceeds the floor exactly in the cells with $\rho/N > 10^{-3}$ (single stage at $T \geq 4$, two stages at $T \geq 8$, four stages at $T \geq 16$) and nowhere else. The null arm is therefore a working queue whose small wait appears where PASTA says it must, and splitting under it costs $(N - 1)h$ at every reported quantile.

Verdict. All three analytical statements survive the full-corpus test. The quantitatively load-bearing one, the pathwise bound of Theorem 1, holds on every message of every window, and its slack is measured rather than assumed: $\gamma = 1.00, 1.23, 1.48, 1.65, 1.67$ at $T = 2, 4, 8, 16, 32 \mu\text{s}$.

Part III

Best practices: implications for HFT systems design

6 The reference instance: KASPAR-HFT and its `fast_send` pipeline

The recommendations in this Part are stated for HFT systems in general, but they are worked out on one concrete system, and the reader should know what that system is. KASPAR-HFT [32–34] is an open-source C++20 trading system for CME futures built on a shared-nothing actor framework, available at <https://github.com/vincent212/kaspar-hft>: every component (socket reader, feed arbitrator, SBE decoder, order book, strategy, order manager, iLink session) is an actor with private state and a mailbox, and components interact only by passing messages. It is at once a production system (MDP3 multicast in, full order books reconstructed order by order, the SHADOW-POV execution algorithm [35] on top, authenticated iLink 3 sessions out) and, with the same actor code driven from recorded packet captures, a deterministic replay simulator in its own right. Part II does not run on that simulator: its measurements are made on a standalone tandem-Lindley model of the pipeline (Section 3.3), driven by packet-arrival timestamps extracted from the same class of capture. The framework supplies the architecture under study and the calibration of h ; it is not in the measurement loop. The `fast_send` paper documents the four extensions that make the actor model usable at HFT latencies: `fast_send` itself, a synchronous delivery in which the sending thread runs the receiver’s handler inline and takes the reply as a value; actor `Groups`, which co-schedule a set of actors on one thread behind a single mailbox; mailbox queue implementations selectable per actor; and a memory pool that removes per-message allocation and its tail. The defining property of `fast_send` is *receiver transparency*: a handler is written once and has no way to tell whether it was invoked synchronously or asynchronously, on which thread, or whether its sender is blocked. That property is what Section 7 builds on.

The reference pipeline of that paper is the design context this paper revisits. It has three sections joined by exactly two cross-thread hops. Two socket-reader actors, one per redundant multicast channel, do nothing but drain their UDP sockets and hand datagrams to a buffer actor by an asynchronous `send`; that hop is asynchronous because a reader that blocked on its callee would drop packets under a burst. From the buffer, arbitration between the A and B feeds, SBE decode, order-book construction, and signal generation run as a single chain of `fast_sends` on one thread, with no queue and no context switch between stages. When the signal decides to act it hands the order to an order-management actor by a second asynchronous `send`, because the iLink write takes microseconds and blocking the decode thread on it would stall the market-data queue. The measured costs behind that layout are Table 4 of the `fast_send` paper: a cross-thread `send` round trip of $\sim 3.4 \mu\text{s}$ (a mailbox wakeup across cores), a same-thread grouped `send` of 90 ns, and a `fast_send` of ~ 30 ns. With a servicing chain of $\sim 7 \mu\text{s}$ measured tick-to-book on live ES, NQ, and ZN feeds, the framework’s own per-message contribution is under 1% of the floor, and the same measurement already found that the latency tail was set by the arrival process, which was strongly clustered, with a Fano-implied branching ratio of 0.85–0.97. That observation is the seed of the present paper.

Read against Part II, the reference pipeline is the single-thread servicing chain of Section 1.2 realised with actors: the whole T sits on one thread between two hops that were accepted because the alternative, blocking the chain, was worse. The two existing hops show that the design was never opposed to asynchronous handoffs as such; it placed them where the cost of not having them was obvious. What this paper adds is that the same logic applies inside the chain. On a Hawkes-clustered

feed the $\sim 7\mu\text{s}$ servicing chain sits above the $2h \approx 3.4\mu\text{s}$ floor of Section 4.3, its single-thread p_{99} carries a tail excess of several T (Table 2), and cutting it at its natural boundaries into stages joined by asynchronous **send** compresses that excess by the factor of Theorem 1 for a median cost of one hop per cut. The recommendations that follow are therefore stated as changes to this particular pipeline: where to cut it, in what order to place the stages, which boundaries to leave on **fast_send**, and how to decide each of those by measurement rather than by reasoning at a single call site.

The rule applied to this pipeline. The design equation can be run on the reference chain directly, using Table 2 as a lookup of the single-server tail excess $\Delta_{\text{single}}(s)$ against per-stage service s . Take the measured $\sim 7\mu\text{s}$ servicing chain and the four natural boundaries inside it (arbitration, SBE decode, book application, signal), and suppose for the sake of the arithmetic that the decode and book stages dominate at roughly $3\mu\text{s}$ each with arbitration and signal near $0.5\mu\text{s}$. The partitions available at those cut points, with $h = 1.7\mu\text{s}$ and Δ_{single} interpolated on the power-of-two grid of Table 2, are:

partition of the $7\mu\text{s}$ chain	s_{max} (μs)	hops	$\Delta_{\text{single}}(s_{\text{max}})$ (μs)	p_{99} excess + hop tax
single thread (7)	7.0	0	≈ 17	17.0
3.5 + 3.5 (decode book)	3.5	1	≈ 5.9	7.6
3 + 3 + 1 (decode book rest)	3.0	2	≈ 4.6	8.0
0.5+3+3+0.5 (all four split)	3.0	3	≈ 4.6	9.7

Three things follow, and each is an instance of a general rule rather than a fact about this pipeline. The balanced two-way cut is the best of the four: it more than halves the tail excess for one hop. Going to three stages lowers s_{max} only from 3.5 to $3.0\mu\text{s}$, because the book stage is already the bottleneck and the third cut merely separates two sub-tasks that were not binding; the $1.3\mu\text{s}$ it buys does not cover the $1.7\mu\text{s}$ hop it costs. Splitting all four is strictly worse again: the fourth hop lowers s_{max} not at all, which is Remark 2.2’s third consequence in concrete form. The SHADOW-POV execution algorithm [35] rides the signal stage of this chain; at the illustrative $\sim 0.5\mu\text{s}$ it is the smallest of the four sub-tasks and sits well below the book-application bottleneck, so the design equation keeps it on **fast_send** with its neighbour rather than on a mailbox hop of its own (Remark 2.2, downstream-of-bottleneck merge) and no cut falls on it. So the recommendation for this pipeline is one cut, at the decode/book boundary, with the book stage placed at ingress if the decode is the smaller of the two — and the numbers above are a prior for what the end-of-build measurement of Section 7.3 should find, not a substitute for running it. The per-stage service times are illustrative; the operator’s own profile of the chain replaces them, and the lookup table does the rest.

7 Implementation on KASPAR-HFT: two dispatch paths, one handler

The central practical claim of this paper is that on Hawkes-clustered feeds the choice of **fast_send** versus asynchronous **send** at each actor boundary is not an early-stage architectural commitment but a late-stage optimization decision — one that should be made per boundary on the basis of measured end-to-end median and tail latency of the assembled pipeline against the target feed, not on the basis of local reasoning at any single call site. KASPAR-HFT was designed to support this workflow. This section states the property that makes it possible, argues why it is critical from a system-optimization standpoint, and gives the operational discipline it enables.

The design equation maps onto framework choices in a specific way. The paper’s model of an N -stage tandem assumes each stage’s actor runs on its own thread and sends its output to the next stage through an asynchronous mailbox — that is what makes the pipeline pipelined, and the hop cost h is the wall-clock latency of that asynchronous handoff.

Two hop paths in the shipping KASPAR-HFT framework. Of the delivery mechanisms described in Section 6, two matter for the design equation. The default asynchronous `send` uses `BQueue`, a blocking mailbox built on `std::mutex + std::condition_variable + boost::circular_buffer`; a producer thread enqueues under the mutex and a consumer thread waits on the condition variable. This is the hop path the paper’s Section 3.3 model corresponds to, and halving its measured $\sim 3.4\mu\text{s}$ round trip gives the one-way hop cost $h \approx 1.7\mu\text{s}$ used throughout Part II.

The `fast_send` path is different: it is a *synchronous* dispatch that acquires a per-actor mutex, calls the receiver’s handler inline on the sender’s thread, and returns the reply. There is no queue and no cross-thread wake-up. Its per-call cost is much smaller than the `BQueue` path’s, but it is not a pipelined hop: the sender blocks until the receiver finishes, so a chain of `fast_sends` serialises through all stages on one thread and offers none of the throughput speedup of Proposition 1. The paper’s design equation does not apply to `fast_send` chains.

7.1 Receiver transparency

Every actor in KASPAR-HFT registers its handlers through the `MESSAGE_HANDLER(MsgType, handler_method)` macro. A handler is a function of one message argument; nothing in its body can access, observe, or condition on the delivery mode by which it was invoked. This property is called *receiver transparency* in Maciejewski [33] and is what makes the send-mode choice a purely sender-side decision.

Concretely:

- **Synchronous delivery:** `target->fast_send(m, this)`. The sender’s thread acquires `target`’s per-actor mutex, calls the receiver’s handler inline on the sender’s thread, and takes the reply as a return value. No queue, no cross-thread wake-up. Round-trip cost $\sim 30\text{ ns}$ per Table 4 of Maciejewski [33].
- **Asynchronous delivery:** `target->send(m, this)`. The message is enqueued on `target`’s `BQueue` mailbox (`mutex + condition_variable + boost::circular_buffer`); `target`’s own thread wakes up on the condition variable and runs the handler. Round-trip cost $\sim 3.4\mu\text{s}$ across threads, or $\sim 90\text{ ns}$ same-thread within a `Group`, per the same reference.

Both paths deliver the same handler with the same message. Neither the handler’s body nor any actor downstream of it can tell which path fired. Switching a boundary between the two is a one-line change at the sender’s call site — `target->fast_send(m, this) ↔ target->send(m, this)` — with no code change on the receiver side, no header change, no interface change, no test change.

7.2 Why measurement, not local reasoning, must decide

The end-to-end latency profile of an HFT pipeline is a joint property of (i) which stages exist and where they are cut, (ii) which pairs of adjacent stages are joined by `fast_send` versus `send`, and (iii) the arrival law of the input feed. Proposition 1 and Theorem 1 both fold across multiple boundaries: the throughput speedup is N -fold and the tail contraction is $N^{-\gamma}$ with $\gamma \geq 1$, so the joint optimum is emphatically not deducible from local reasoning at any single boundary. A designer who reasons at one boundary at a time — "this hop is $\sim 1.7\mu\text{s}$, that is a lot, use `fast_send`" —

systematically underestimates the aggregate tail benefit of adding hops, and this is exactly the class of local reasoning that led the low-latency-systems community into the single-thread orthodoxy of Section 1.2.

Getting the joint optimum therefore requires end-to-end measurement of the assembled pipeline against a representative feed, and iteration over per-boundary flips. The observable objective is a two-tuple (p_{50}, p_{99}) of end-to-end latency under the target feed, with an operator-chosen tradeoff between the two (Section 4.3’s κ). The design equation of Part II supplies a first-principles prior on where the optimum sits under a Hawkes feed, but the measurement is the arbiter, not the equation.

7.3 Operational discipline

The KASPAR-HFT design flow that this suggests, and which we now recommend as canonical for HFT actor pipelines running on self-exciting feeds, is:

1. Draft the pipeline as a chain of actors at the natural cut points of the servicing chain (`SocketReader`, SBE decode, order book, strategy, order manager on CME MDP3; four to five stages, with no artificial subdivision of any one of them).
2. Wire every intra-servicing-chain boundary with whichever of `fast_send` or `send` is convenient during development; do not commit to a topology at this stage.
3. Once the pipeline runs end-to-end on the target feed, measure the corpus-median p_{50} and p_{99} of end-to-end latency under the operator’s live-representative tape, and record them as the baseline.
4. Perturb one boundary at a time — flip its `fast_send` $\leftrightarrow$ `send` — and re-measure. Accept the flip if (p_{50}, p_{99}) improves under the operator’s chosen tradeoff.
5. Iterate until no single-boundary flip improves the objective.

Under the arrival law of a CME MDP3 feed the design equation of Section 4.3 predicts what steps 4–5 converge to: `async send` at every boundary whose removal would raise the largest stage, and `fast_send` everywhere else, in particular between short sub-tasks below the $2h$ floor of Section 4.3 and between sub-tasks downstream of the bottleneck whose combined service stays at or below $s_{\max}$ (Remark 2.2). The operator does not need to trust that prediction to run the discipline; the measurement is the arbiter, and the arrival-law prior tells the operator where to expect it to land.

7.4 Separation of code construction from latency optimization

A consequence of Section 7.3 that deserves separate emphasis: in KASPAR-HFT *two* distinct topology decisions are not knowable at design time. The first is the *actor-to-thread mapping* — which actors share a thread by being placed in a common `Group`, and which run on their own thread. The second is the *per-boundary send-mode choice* — whether each pair of adjacent actors is joined by synchronous `fast_send` or asynchronous `send`. Neither decision is deducible from the source code alone. Both are functions of the target feed’s arrival statistics and the operator’s median-versus-tail tradeoff, both of which are properties of the deployment environment. The optimum can only be measured from the assembled system running against a representative feed.

Under receiver transparency this is a feature rather than a limitation. Actor code, message-type headers, and the full suite of unit and integration tests can be written and run before either the actor-to-thread mapping or the send-mode choice has been decided. Code construction and testing therefore proceed independently of the latency-optimization consideration — both decisions are

deferred to the end of the build, informed by measurements that could not have been taken any earlier because the system did not yet exist to measure. Correctness is established first; latency is optimized against the specific feed the system will run on, second. Two developers building two actors on the two sides of a boundary do not need to agree on whether their actors will ultimately live on the same thread or on different threads, nor on whether their handoff will be synchronous or asynchronous; they need to agree only on the message type. The message-type agreement is checkable at compile time; the two topology decisions are scalar dials adjusted at deployment.

This decoupling of correctness-phase engineering from latency-phase engineering is the practical payoff of receiver transparency, and it is what makes KASPAR-HFT’s design cycle qualitatively different from frameworks in which either topology decision leaks into the receiver’s interface (Section 7.5). Design proceeds without a commitment to a topology; the commitment is made once, near deployment, on the basis of measurements the code did not need to know about while it was being written.

7.5 Contrast with non-fungible frameworks

Most low-latency actor frameworks bake the sender’s choice of delivery mode into the receiver’s type. An Akka actor’s `tell` versus `ask`, a CAF actor’s `send` versus `request`, and an Erlang process’s `cast` versus `call` each expose different receiver-side surfaces, so flipping a boundary from sync to async is a small refactor rather than a one-line change. Under those frameworks the design flow of Section 7.3 is not tractable, and the operator’s incentive is therefore to lock in the send-mode choice early — reasoning locally — rather than to defer it to an end-of-build measurement pass. This structural incentive is a plausible contributing cause of the single-thread orthodoxy: frameworks that penalise deferral push designers toward the topology that is least sensitive to a wrong early guess, which is the collapsed single-thread servicing chain.

KASPAR-HFT’s joint mailbox API of `fast_send` and `send`, and the receiver-transparency property underneath, are the concrete machinery that make measurement-driven send-mode optimization practical. That is the framework property this paper’s design rule depends on operationally, and it is why we regard it as critical to system-level optimization on Hawkes-clustered feeds.

8 Other architectural considerations to re-evaluate under Hawkes

The paper’s central claim is that the single-thread hot-path recommendation, treated as received wisdom in HFT infrastructure for over a decade, becomes wrong once the arrival law is recognised as Hawkes-clustered rather than Poisson (Section 1.2, Section 4). Two adjacent architectural conventions in the same low-latency-systems canon — pre-spinning receivers and pre-warming pipelines — rest on the same implicit Poisson-load assumption, and this section flags them as deserving the same treatment. We do not measure either intervention here; both belong to future work. What follows is an argument that the arrival law itself provides most of the benefit these interventions are engineered to deliver, on the specific quantile that HFT operators care about.

Pre-spinning receivers. The busy-poll recommendation of Rigtorp [39], Belay et al. [61], and the Aeron/Chronicle lineage (Section 1.2) pins a spinning thread per receiver so the wake path from a sleeping condition variable is never taken. On a Poisson feed, any arrival can find the receiver asleep, and the wake latency is paid on average; pre-spinning eliminates it. On a Hawkes feed with branching ratio $n \approx 0.8$, sleep/wake transitions happen only at cluster boundaries. Every p_{99} message is intra-cluster (Section 4.1: the p_{99} message waits behind a cluster, since under the Poisson null it

does not wait at all), so by the time a tail-setting message arrives, the receiver has been continuously processing for many multiples of the per-message service time and is already effectively spinning as a natural consequence of the load. Under the model used here, pre-spinning therefore saves little or no wake latency at the tail; this is a prediction of the queueing model, not a measurement of a production receiver’s state, and should be evaluated directly. It saves wake latency for the median and for the minimum-latency floor (inter-cluster arrivals which do find the receiver asleep after seconds-scale gaps), but the saving is bounded by $\sim 1.4 \mu\text{s}$ per hop (the $1.7 \mu\text{s}$ wake-path cost less the few-hundred-nanosecond mutex and cache-line floor of Section 4.5) and is much smaller than the tail win from pipelining. There is also an operational cost: in the present author’s production experience, pinning spinning threads across the servicing chain has caused overall system degradation — steady core burn at 100%, cache thrash between the spin loop and the actor’s own working set, and pressure on shared resources like the LLC and memory bandwidth — that appears to eat back the modest median-latency gains, though this is anecdotal and not measured here.

Pre-warming. Pre-warming is arguably not a separate consideration but a subset of the wake-up cost already discussed above. The $h = 1.7 \mu\text{s}$ one-way hop cost that this paper calibrates already captures the entire process-wake-up path on the receiver side: the kernel context switch, the reload of the receiver’s instruction cache, the L1/L2 misses on its actor state and message pool, the TLB refill, and the scheduler putting the thread back on a core. To the extent the thread hop is dominated by cluster-wait on Hawkes feeds — because every p_{99} arrival is intra-cluster and the receiver is, under the model, already awake — the cache-warmth components inside that h should follow the same pattern, since they are the same h decomposed. This is a consequence of the model rather than a measured result. Pre-warming interventions that fire synthetic traffic to keep instruction caches, data caches, TLBs, and branch predictors hot are therefore attacking the same h that Section 4 finds is dominated by cluster-wait. What such interventions can meaningfully deliver is confined to the median and the minimum — the inter-cluster first-arrival case where the receiver truly did cool down — and even there the delivery is bounded by the wake-up cost itself and comes at the price of continuous L1/L2 pollution from the warmup traffic. Under Hawkes the tail sees none of this.

Why this needs measurement. The received wisdom that treats pre-spinning and pre-warming as first-order tail-latency techniques is, like the single-thread rule, a Poisson-load argument in disguise. On real self-exciting exchange feeds the arrival law itself provides the receiver-alive and cache-warm state on the quantile that matters (the tail), so both interventions are second-order at best on those quantiles and can be actively counterproductive once their steady costs are accounted for. Both claims are corollaries of the paper’s main result rather than separate contributions, but each deserves a dedicated empirical evaluation on Hawkes-clustered feeds, analogous to the single-thread evaluation this paper carries out. The two open questions are:

- What is the actual ROI of pinned busy-poll receivers on the Hawkes-load $p_{50}, p_{99}, \text{max}$ triple, net of the steady core-burn cost and the cache pressure the spin loop imposes on the actor’s own working set?
- What is the actual ROI of periodic pre-warming traffic on the same triple, net of the L1/L2 pollution it creates and the residual advantage it delivers only on inter-cluster first-arrivals?

Both are direct extensions of the present paper’s methodology (measure under real CME MDP3 tapes; use a matched Poisson null to isolate the arrival-law dependence) and both are candidate follow-ons.

9 Recommendations for HFT system designers

The paper’s results overturn one architectural default (single-thread hot path) and cast serious doubt on two adjacent ones (busy-poll receivers, pre-warming). It is easy to read those results as an attack on the microsecond-shaving craft that HFT infrastructure engineers spend most of their time on, and they are not. This section separates what the paper leaves untouched from what it argues against, and states the design flow the paper’s design equation actually implies.

9.1 What the paper leaves untouched

The bit-level hot-path optimization craft — branch-free decoders, cache-line-aligned actor state, atomic-free single-writer queues, SBE decode in place, memory pools, `fast_send` at ~ 30 ns, kernel-bypass NICs (Solarflare Onload, DPDK, VMA), `isolcpus`, `SCHED_FIFO`, huge pages, RTC counters instead of `clock_gettime`, non-blocking IO on the socket, avoidance of allocation on the hot path, careful use of the branch predictor — is every bit as necessary as it was before this paper. All of it shows up in the service floor T , which the paper takes as given and treats as the constant the multi-actor pipeline is compressing the tail *around*. The lower T is, the better the pipelining rule performs: Proposition 1 gives an N -factor throughput speedup at any T , and Theorem 1’s $1/N$ bound applies to a single-stage wait $\Delta(1)$ that itself scales with T . So the pipelining rule is complementary to, not a substitute for, the microsecond-shaving craft: a $5\mu\text{s}$ -servicing-chain implementation of a two-stage tandem beats a $10\mu\text{s}$ -servicing-chain implementation of the same two-stage tandem at every quantile, and the difference is entirely in the fast-code layer this paper does not touch.

9.2 What the paper does argue against

Two layers above the bit-level craft, both of which have been received wisdom in the HFT community for a decade or more:

1. **Topology default: single-thread servicing chain.** The Poisson-orthodox recommendation to collapse the entire hot path onto one thread (Section 1.2). On Hawkes-clustered feeds this is empirically wrong: the tail is dominated by cluster-wait, and adding threads between actors along the servicing chain compresses that tail by $N^{-\gamma}$ at a small $(N - 1)h$ median cost, provided T sits above the $2h$ floor of Section 4.3.
2. **Always-on interventions: pre-spinning receivers and pre-warming pipelines.** The Poisson-orthodox recommendation to burn a core-worth of CPU per receiver on busy-poll, and a smaller amount of CPU on periodic warmup traffic, to eliminate wake-up latency and cold-miss cost on the receiver’s side (Section 8). On Hawkes-clustered feeds the receiver is already effectively spinning and effectively warm at the arrival time of every p_{99} message, so these interventions target quantiles (median, minimum) that are not what an HFT operator cares about, and they pay a steady core-burn and cache-pollution cost for it.

These are orthodoxy about the *layer above* the fast-code craft, not about the fast-code craft itself. Practitioners who conflated “I made this loop 200 ns faster” with “therefore single-thread wins” were making two different claims at once; the paper only overturns the second.

9.3 Recommended design flow

The design flow that follows from the paper’s results, and that we recommend as the canonical HFT infrastructure workflow on self-exciting exchange feeds:

1. **Keep the fast-code craft.** Everything in the first subsection above. This is orthogonal to the paper’s recommendation and is still what pays the bills on absolute latency.
2. **Draft the pipeline as a chain of actors at natural cut points, then balance it.** On a CME MDP3 handler: `SocketReader`, SBE decoder, order book, strategy actor, order manager — four or five stages, no artificial decomposition (Section 7). The tail factor is $s_{\max}/T$ (Remark 2.2), so the goal of the decomposition is the smallest largest stage, and the largest stage belongs at ingress so that all queueing sits in one mailbox (Remark 2.2).
3. **Wire boundaries with whichever of `fast_send` or `send` is convenient during development.** Receiver transparency in KASPAR-HFT makes this a one-line-per-boundary decision that can be deferred (Section 7).
4. **Do not commit to an actor-to-thread mapping at design time.** Grouping decisions and per-boundary send-mode decisions are both scalar dials at deployment, not code-structure decisions (Section 7.4).
5. **Measure the assembled pipeline against a representative feed.** Corpus-median p_{50} and p_{99} of end-to-end latency, per operator-chosen tradeoff (Section 7.3).
6. **Flip one boundary at a time and re-measure.** Accept the flip if (p_{50}, p_{99}) improves. Iterate until no single-boundary flip helps.
7. **Prior on where step 6 converges.** On Hawkes-clustered feeds, async `send` at every boundary whose removal would raise the largest stage; `fast_send` at every other boundary, in particular between sub-tasks downstream of the bottleneck whose combined service stays at or below $s_{\max}$, and between short sub-tasks below the $2h$ floor. A hop that does not lower $s_{\max}$ costs h at every quantile and compresses nothing.
8. **Do not pre-spin or pre-warm by default.** If pre-spinning or pre-warming is used, measure the ROI on a Hawkes-representative tape, net of the steady core-burn and cache-pollution costs. If the measurement improves the tail materially, keep the intervention. The paper’s prior is that on Hawkes feeds it will not (Section 8).

The workflow prioritises measurement over local reasoning at every step, on the specific grounds that the arrival law of the target feed is a first-order determinant of the optimum and is not a property of the code that can be introspected from the sender’s call site.

Part IV

Conclusions

10 Conclusion

The paper’s two headline conclusions overturn the received wisdom in HFT system design. That wisdom holds that the entire hot path should live on a single-thread event loop, on the grounds that every thread hop adds real per-message latency; the community is overwhelmingly in the single-thread camp. We show on real CME MDP3 packet flow that both halves of the argument fail.

Conclusion 1: the thread-hop cost is a wash under real arrivals. The dominant term in end-to-end latency on a Hawkes-clustered feed is not the mailbox handoff — it is the cluster-wait tail. On 3512 windows at $h = 1.7 \mu\text{s}$ the corpus-median single-thread p_{99} is 2.7, 3.5, 5.0, and 8.0 times the service floor at $T = 4, 8, 16, 32 \mu\text{s}$, i.e. a tail excess of 4 to 130 times the hop cost; the $p_{99.9}$ excess is twice that again. Adding a hop is $+1.7 \mu\text{s}$ at every quantile; the tail it lets us compress is one to two orders of magnitude larger whenever T sits above the $2h$ floor. The single-thread objection — “every hop is a fixed tax that no benefit can overcome” — is empirically wrong on this class of feed.

Conclusion 2: two threads beat one on the tail at the cost of one hop on the median. Splitting a service task into two half-length actors on separate threads joined by one $1.7 \mu\text{s}$ mailbox hop lowers p_{99} on 100% of the 3512 windows at every service floor from 4 to $32 \mu\text{s}$, cutting the corpus-median tail excess to between 0.28 and 0.34 of its single-thread value for a median cost of exactly h ; the trade favours the split for any tail weight $\kappa \gtrsim 0.35$ at $T = 4 \mu\text{s}$ and $\kappa \gtrsim 0.01$ at $T = 32 \mu\text{s}$. At $T = 2 \mu\text{s}$, below the $2h$ floor of Section 4.3, the split loses on p_{99} , as the design equation says it must. The pattern extends to $N > 2$: an N -stage tandem contracts the tail excess by a factor of N or better on every sample path (Theorem 1) while paying $(N - 1)h$ at every quantile, and the p_{99} -minimising stage count rises with the task length, from 2 at $4 \mu\text{s}$ to 8 at $32 \mu\text{s}$, up to the number of natural cut points in the decode chain (four or five on CME). The single-thread event loop is dominated on the tail on Hawkes-clustered feeds above the $2h$ floor, and holds its advantage only on the median, by one hop per stage.

Why the received wisdom persists. We think the field arrived at the single-thread rule under a mental model of Poisson-like arrivals, where clustering does not exist and the tail sits near the service floor. On such a feed the received wisdom is correct — our own results give exactly that answer under the Poisson null. What the field missed is that real exchange feeds have never been Poisson: near-critical branching ratios on CME futures ($n \approx 0.9$ on ES trades in the literature, $n \approx 0.8$ on NQ packets in this corpus) have been documented for over a decade, and it is precisely those clusters that reverse the architectural recommendation. When your feed is bursty, thread-hop overhead is the price of admission to an operating point with a far shorter tail at a bounded median cost.

What this changes in practice. Multi-actor pipelines are the correct architecture for HFT feeds specifically because those feeds are self-exciting. Framework designers should measure busy-poll dequeue on pinned isolated cores against the wait/notify mailbox on a Hawkes-representative tape rather than assume it wins (Section 8), drive the mailbox mutex uncontended, keep memory-pool discipline, and cut the decode pipeline at whichever of its natural boundaries (4–5 on CME) minimise the largest stage, with that stage at ingress and everything downstream merged up to its service

time; the tail factor is $s_{\max}/T$, and a hop that does not lower $s_{\max}$ is pure cost. Strategy designers of both POV-style and race-sensitive algorithms benefit from the same N budget for the same reason: the tail is what matters, and the pipeline compresses it.

Retraction of a prior design context, and the corrected design rule. An earlier paper by the present author [33] did not itself argue for the single-thread servicing chain; it took the collapsed single-thread servicing chain as the widely accepted design context of the time and offered `fast_send` as the mechanism that makes actor composition inside it free of context-switch overhead. What the present paper retracts is the design context, not the mechanism: on Hawkes-clustered feeds the single-thread servicing chain should no longer be the unconditional default. The corrected position is that `fast_send` (synchronous, inline, sub-microsecond) and asynchronous `BQueue` handoffs (a real pipelined hop at a few microseconds) are *complementary* tools, and the choice between them is a per-boundary design decision that depends on the service time being chained and the arrival law of the feed the pipeline sits on. Designers should use `fast_send` and asynchronous sends judiciously, according to which is appropriate for the task at hand.

In shipping KASPAR-HFT the `fast_send` vs asynchronous-send choice at each actor boundary is normally left open until the end of the build, and is resolved by measuring the end-to-end median and tail latency of the assembled pipeline under the target feed. A convenient property of the KASPAR-HFT actor API is that the receiving actor’s `MESSAGE_HANDLER` is agnostic to how it was invoked: swapping a synchronous `fast_send` at a call site for an asynchronous `send` (or vice versa) is a one-line change on the sender side and requires no code change on the receiver side. The choice therefore lives entirely at each individual call site and can be flipped per boundary at end-of-build with minimal engineering cost. The paper’s design equation now supplies the arrival-law-aware default value of that end-of-build measurement: on Hawkes-clustered feeds, long tasks should be split into shorter actors joined by asynchronous sends as a matter of preference, on the presumption that splitting compresses the overall tail more than the added hop cost inflates the median. The rule presupposes that the task being split has a service time T comfortably above the $2h$ floor of Section 4.3; a task of order h or below cannot be usefully split (a $1\mu\text{s}$ task cannot be turned into two 500 ns actors when a single hop alone costs $1.7\mu\text{s}$). Short synchronous chains inside one actor’s servicing of one message therefore continue to use `fast_send` unchanged. The end-of-build measurement remains the authoritative arbiter of any specific boundary.

Main caveats. The corpus is a single instrument (NQ front-month, chan 318); ES and BTC replications are follow-on work. Constant service is a simplification; the true per-actor work has state-dependent variability we do not model here. The cache-pollution cost of spinning receivers on many cores is bounded but not measured. And Theorem 1 is a bound whose slack depends on the arrival law; the paper measures that slack (γ between 1.0 and 1.7 against the guaranteed $\gamma \geq 1$) rather than deriving it, and a closed form for the slack on Hawkes input is open.

None of these caveats reverse the two conclusions on the 276-session corpus.

11 Future work

- Multi-instrument replication (ES, BTC).
- Closed form for the slack in Theorem 1 under Hawkes input, i.e. the exponent $\gamma - 1$ as a function of $(\rho, n, \beta T)$. By the reduction this is a single-server question about $M^{[K]}/D/1$ with Borel batches at two service levels, which is tractable; the paper measures γ instead.

- Cache-pollution empirical measurement per spinning-receiver stage count.
- Power-law-kernel Hawkes fit versus exponential.
- Overnight-session regime.
- Own-capture pcap validation of the network sub-stage.
- Load-conditioned hop cost $h(\rho)$. This paper calibrates $h = 1.7 \mu\text{s}$ from the low-load wake-latency figure in Maciejewski [33]; under sustained-load conditions inside a Hawkes cluster the receiver never sleeps and the effective h collapses toward the mutex + cache-line-transfer floor of a few hundred nanoseconds (Section 4.5). Measuring $h(\rho)$ directly and re-running the sweep against the load-conditioned h would sharpen the reported tail-win, which the current single- h sim understates.

12 Reproducibility

All code referenced below lives in the KASPAR-HFT repository [34], <https://github.com/vincent212/kaspar-hft>, which also carries the actor framework the paper’s recommendations are stated for.

Code: `arrival_paper/qsim_prep.py` (stage 1: per-window packet-arrival extraction and packet-Hawkes fit, cached to `metadata.parquet` plus per-window arrival arrays) and `arrival_paper/qsim_run.py` (stage 2: the (T, h, N) tandem Lindley grid with the Poisson null and the per-window Theorem 1 bound check, seconds per full-corpus sweep under `numba`); `arrival_paper/hawkes_smoke.py` holds the Hawkes MLE. Data: 281 per-session message tapes derived from CME MDP3 channel 318 pcaps (NQ front month, 2025-01 through 2026-02); the tapes are derived from licensed exchange data and are not redistributed, but the extraction script and the cached per-window metadata are released with the paper.

References

- [1] Hawkes, A. G. (1971). Spectra of some self-exciting and mutually exciting point processes. *Biometrika* 58, 83–90.
- [2] Hawkes, A. G., Oakes, D. (1974). A cluster process representation of a self-exciting process. *Journal of Applied Probability* 11(3), 493–503.
- [3] Tanner, J. C. (1961). A derivation of the Borel distribution. *Biometrika* 48(1–2), 222–224.
- [4] Asmussen, S. (2003). *Applied Probability and Queues*, 2nd ed. Springer. Chapter X (heavy-tailed and subexponential asymptotics for single-server queues).
- [5] Gao, X., Zhu, L. (2018). Functional central limit theorems for stationary Hawkes processes and application to infinite-server queues. *Queueing Systems* 90, 161–206.
- [6] Jackson, J. R. (1957). Networks of waiting lines. *Operations Research* 5, 518–521.
- [7] Burke, P. J. (1956). The output of a queuing system. *Operations Research* 4, 699–704.
- [8] Kelly, F. P. (1979). *Reversibility and Stochastic Networks*. Wiley.
- [9] Kleinrock, L. (1976). *Queueing Systems, Vol. II: Computer Applications*. Wiley.

- [10] Whitt, W. (1983). The queueing network analyzer. *Bell System Technical Journal* 62, 2779–2815.
- [11] Friedman, H. D. (1965). Reduction methods for tandem queueing systems. *Operations Research* 13(1), 121–131.
- [12] Lindley, D. V. (1952). The theory of queues with a single server. *Mathematical Proceedings of the Cambridge Philosophical Society* 48(2), 277–289.
- [13] Foss, S., Korshunov, D. (2000). Steady-state asymptotics for tandem, split-match and other feedforward queues with heavy-tailed service. *Queueing Systems* 35, 65–80.
- [14] Baccelli, F., Foss, S., Lelarge, M. (2005). Tail asymptotics for monotone-separable networks. *J. Appl. Probability* 42(3), 793–812.
- [15] Heindl, A. (2001). Decomposition of general tandem queueing networks with MMPP input. *Performance Evaluation* 44(1–4), 5–23.
- [16] Kim, Y. H., Chydzinski, A. (2002). Connection-wise end-to-end performance analysis of queueing networks with MMPP inputs. *Performance Evaluation* 47(2–3), 137–156.
- [17] Dębicki, K., Mandjes, M. (2015). *Queues and Lévy Fluctuation Theory*. Springer.
- [18] Daw, A., Pender, J. (2018). Queues driven by Hawkes processes. *Stochastic Systems* 8(3), 192–229.
- [19] Koops, D. T., Boxma, O. J., Mandjes, M. R. H. (2018). Infinite-server queues with Hawkes arrivals. *Queueing Systems* 92, 27–52.
- [20] Chen, X., Blanchet, J. (2021). Perfect sampling of Hawkes processes and queues with Hawkes arrivals. *Stochastic Systems* 11(3), 264–283.
- [21] Sheldon, D., et al. (2023). Steady-state analysis and online learning for queues with Hawkes arrivals. arXiv 2311.02577.
- [22] Gao, X., Zhu, L. (2024). Single-server queues with state-dependent Hawkes arrivals. *Mathematics of Operations Research*.
- [23] Ozaki, T. (1979). Maximum likelihood estimation of Hawkes’ self-exciting point processes. *Ann. Inst. Statist. Math.* 31(1), 145–155.
- [24] Filimonov, V., Sornette, D. (2012). Quantifying reflexivity in financial markets: toward a prediction of flash crashes. *Physical Review E* 85, 056108.
- [25] Hardiman, S. J., Bercot, N., Bouchaud, J.-P. (2013). Critical reflexivity in financial markets: a Hawkes process analysis. arXiv 1302.1405.
- [26] Bacry, E., Mastromatteo, I., Muzy, J.-F. (2015). Hawkes processes in finance. *Market Microstructure and Liquidity* 1(1).
- [27] Byrd, D., Hybinette, M., Balch, T. H. (2020). ABIDES: Towards high-fidelity market simulation for AI research.
- [28] Frey, S., Li, K., Nagy, P., Sapora, S., Lu, C., Zohren, S., Foerster, J., Calinescu, A. (2023). JAX-LOB. arXiv 2308.13289.

- [29] Aquilina, M., Budish, E., O’Neill, P. (2022). Quantifying the high-frequency trading “arms race”. *Quarterly Journal of Economics* 137(1), 493–564.
- [30] Dean, J., Barroso, L. A. (2013). The tail at scale. *Communications of the ACM* 56(2), 74–80.
- [31] Zhao, K., Alizadeh, M., Pan, R., Ports, D. R. K., Perry, J., Yang, X., et al. (2023). Parsimon: scalable tail latency estimation for data center networks. NSDI 2023.
- [32] Maciejewski, V. (in preparation). KASPAR-HFT: a shared-nothing actor framework for CME futures — `fast_send` hop-cost measurement.
- [33] Maciejewski, V. (2025). `fast_send`: synchronous inline dispatch on the CME MDP3 hot path. M2 Tech technical report, KASPAR-HFT repository. https://github.com/vincent212/kaspar-hft/blob/main/tech_reports/fast_send.pdf
- [34] Maciejewski, V. (2026). KASPAR-HFT: an open-source C++20 actor framework and trading system for CME futures. Source repository. <https://github.com/vincent212/kaspar-hft>
- [35] Maciejewski, V. (2026). Model-free passive execution via order-level shadowing. arXiv:2609.18019. <https://arxiv.org/abs/2609.18019>
- [36] Thompson, M. (2011). The Single Writer Principle. *Mechanical Sympathy*, 29 September 2011. <https://mechanical-sympathy.blogspot.com/2011/09/single-writer-principle.html>
- [37] Fowler, M. (2011). The LMAX Architecture. 12 July 2011. <https://martinfowler.com/articles/lmax.html>
- [38] LMAX Exchange (2011). Disruptor: high-performance alternative to bounded queues for exchanging data between concurrent threads. <https://lmax-exchange.github.io/disruptor/>
- [39] Rigtorp, E. (2020). Low latency tuning guide. <https://rigtorp.se/low-latency-guide/>
- [40] Chronicle Software (2023). Building fast trading engines: Chronicle’s approach to low-latency trading. <https://chronicle.software/insights/blogs/building-fast-trading-engines-chronicles-approach-to-low-latency-trading>
- [41] Real Logic (2016). Aeron: efficient reliable UDP unicast, UDP multicast, and IPC message transport. <https://aeron.io/>
- [42] Kwan, A., Wang, H., Luk, W. (2023). C++ design patterns for low-latency applications including high-frequency trading. arXiv 2309.04259.
- [43] Hu, W. (2024). The deterministic event-driven sequencer architecture: a competitive edge for high-frequency trading. Medium, October 2024. <https://medium.com/@hu.wenzhe124124/the-deterministic-event-driven-sequencer-architecture-a-competitive-edge-for-high-frequency>
- [44] Schmidt, D. C. (1995). Reactor: an object behavioral pattern for concurrent event demultiplexing and event handler dispatching. In *Pattern Languages of Program Design*, J. O. Coplien and D. C. Schmidt (eds.), Addison-Wesley.
- [45] Kegel, D. (1999). The C10K problem. <https://www.kegel.com/c10k.html>
- [46] Sysoev, I. (2004). nginx: a high-performance HTTP server and reverse proxy. <https://nginx.org/>

- [47] Dahl, R. (2009). Node.js: JavaScript runtime built on the V8 engine. Presented at JSConf EU, Berlin, November 2009. <https://nodejs.org/>
- [48] Welsh, M., Culler, D., Brewer, E. (2001). SEDA: an architecture for well-conditioned, scalable internet services. *Proc. 18th ACM Symposium on Operating Systems Principles (SOSP)*, 230–243.
- [49] Welsh, M. (2010). A retrospective on SEDA. <https://matt-welsh.blogspot.com/2010/07/retrospective-on-seda.html>
- [50] Kohler, E., Morris, R., Chen, B., Jannotti, J., Kaashoek, M. F. (2000). The Click modular router. *ACM Transactions on Computer Systems* 18(3), 263–297.
- [51] Murray, D. G., McSherry, F., Isaacs, R., Isard, M., Barham, P., Abadi, M. (2013). Naiad: a timely dataflow system. *Proc. 24th ACM Symposium on Operating Systems Principles (SOSP)*, 439–455.
- [52] Marty, M., de Kruijf, M., Adriaens, J., Alfeld, C., et al. (2019). Snap: a microkernel approach to host networking. *Proc. 27th ACM Symposium on Operating Systems Principles (SOSP)*, 399–413.
- [53] Kaffes, K., Chong, T., Humphries, J. T., Belay, A., Mazières, D., Kozyrakis, C. (2019). Shinjuku: preemptive scheduling for μ second-scale tail latency. *Proc. 16th USENIX Symposium on Networked Systems Design and Implementation (NSDI)*, 345–359.
- [54] Ousterhout, A., Fried, J., Behrens, J., Belay, A., Balakrishnan, H. (2019). Shenango: achieving high CPU efficiency for latency-sensitive datacenter workloads. *Proc. 16th USENIX NSDI*, 361–378.
- [55] Fried, J., Ruan, Z., Ousterhout, A., Belay, A. (2020). Caladan: mitigating interference at microsecond timescales. *Proc. 14th USENIX OSDI*, 281–297.
- [56] Chen, H., Delimitrou, C., Kozyrakis, C. (2021). Perséphone: a microsecond-scale time-sharing scheduler for wide service-time distributions. *Proc. 28th ACM SOSP*, 653–668.
- [57] DPDK Project (2024). DPDK Programmer’s Guide: run-to-completion and pipeline modes. https://doc.dpdk.org/guides/prog_guide/
- [58] Zhang, X. (2016). Pipelined FAST protocol market-data decoder for high-frequency trading. Chinese patent CN105654383A.
- [59] Vinnikov, A. (2014). Batch latency analysis and phase transitions for a tandem of queues. arXiv 1403.2400.
- [60] David, T., Guerraoui, R., Trigonakis, V. (2013). Everything you always wanted to know about synchronization but were afraid to ask. *Proc. 24th ACM Symposium on Operating Systems Principles (SOSP)*, 33–48.
- [61] Belay, A., Prekas, G., Klimovic, A., Grossman, S., Kozyrakis, C., Bugnion, E. (2014). IX: a protected dataplane operating system for high throughput and low latency. *Proc. 11th USENIX Symposium on Operating Systems Design and Implementation (OSDI)*, 49–65.

Appendices

A The Borel batch-size tail at fixed branching ratio

This appendix records the calculation behind Equation (8) and the numerical check that the critical $k^{-1/2}$ survival tail of the Borel distribution is confined to a finite window at every $n < 1$.

Stirling's formula $k! = \sqrt{2\pi k} k^k e^{-k} (1 + O(1/k))$ in Equation (7) gives

$$\Pr(K = k) = \frac{(nk)^{k-1} e^{-nk}}{k!} = \frac{n^{k-1} k^{k-1} e^{-nk}}{\sqrt{2\pi k} k^k e^{-k}} (1 + O(1/k)) = \frac{1}{n\sqrt{2\pi}} k^{-3/2} (ne^{1-n})^k (1 + O(1/k)).$$

Writing $n = 1 - \epsilon$, $\log(ne^{1-n}) = \log(1 - \epsilon) + \epsilon = -\epsilon^2/2 - \epsilon^3/3 - \dots$, so the geometric factor is $\exp(-k\epsilon^2/2(1 + O(\epsilon)))$. The pmf is therefore $\propto k^{-3/2}$ for $k \ll k^* = \epsilon^{-2}$ and geometric beyond; summing, $\Pr(K > k) \asymp k^{-1/2}$ on the same window and $\asymp k^{-3/2} e^{-k\epsilon^2/2}$ beyond it. At $n = 1$ exactly, $k^* = \infty$, $\Pr(K > k) \sim \sqrt{2/\pi} k^{-1/2}$ (Borel–Tanner), and $E[K] = \infty$; that is the unstable case $\rho = \infty$, which no stationary queue attains.

Because the tail is geometric, K has finite exponential moments $E[e^{\theta K}] < \infty$ for $\theta < \epsilon^2/2(1 + O(\epsilon))$ and is therefore outside the subexponential class $\mathcal{S}$. Heavy-tail queueing results that assume subexponential batch work, including the single-big-batch principle for $M^{[K]}/G/1$ and the monotone-separable tail theorem of Baccelli et al. [14], do not apply, and the stationary waiting time of the batch-arrival queue has a Cramér–Lundberg (exponential) tail at every fixed $n < 1$.

Table 3 reports the local log-log slope of the exact pmf, $\Delta \log \Pr(K = k) / \Delta \log k$ between successive k , which a pure $k^{-3/2}$ law would hold at -1.5 .

n	k^*	$10 \rightarrow 30$	$30 \rightarrow 100$	$100 \rightarrow 300$	$300 \rightarrow 1000$	$1000 \rightarrow 3000$
0.80	25	−1.92	−2.84	−5.71	−14.96	−43.63
0.90	100	−1.59	−1.81	−2.48	−4.62	−11.26
0.95	400	−1.52	−1.57	−1.73	−2.25	−3.85
0.99	10000	−1.50	−1.50	−1.51	−1.53	−1.59

Table 3. Local log-log slope of the Borel pmf, Equation (7), between successive cluster sizes k . The critical slope -1.5 holds while $k \ll k^* = (1 - n)^{-2}$ and steepens geometrically beyond.

The practical reading is that at the branching ratios fitted on the corpus ($n_{\text{pkt}} \approx 0.8$, Section 3.2), $k^* \approx 25$: clusters of order a hundred packets are the largest that occur with non-negligible frequency, and the single-stage wait behind the largest clusters has scale $Tk^* \approx 25T$, against the $1.0\text{--}7.0T$ tail excess observed at the corpus-median p_{99} of Section 4 (rising monotonically with T across the sweep). This is the object that Theorem 1 divides by N , and the reason the $n = 0.80$ row of Table 3 is the relevant one.

B Numerical check of Theorem 1

A direct simulation of the model of Theorem 1 checks both the identity Equation (4) and the bound Equation (6). Immigrants arrive as a Poisson process at rate $\mu = 0.004 \mu\text{s}^{-1}$, each spawning a Galton–Watson cluster with Poisson(0.9) offspring, so $K \sim \text{Borel}(0.9)$ and $E[K] \approx 10$; clusters are placed as instantaneous batches, giving $\rho = \bar{\lambda}T \approx 0.40$ at $T = 10 \mu\text{s}$. The tandem has $N = 4$ equal stages of $2.5 \mu\text{s}$ and $h = 1.7 \mu\text{s}$; 2×10^5 messages are simulated with an explicit per-stage FIFO recursion

(independent of the paper’s `qsim_grid.py`). Over all messages, $\max_j |W_j^{(4)} - (w_j(T/4) + 3h)| < 10^{-9} \mu\text{s}$ and Equation (6) is violated by no message. The quantiles are

q	$W_q^{(1)}$	$W_q^{(4)}$	$W_q^{(1)}/4 + 3h$	$(W_q^{(4)} - 3h)/W_q^{(1)}$
0.5	350.0	55.1	92.6	0.14
0.9	2066.6	355.1	521.8	0.17
0.99	5342.1	1127.6	1340.6	0.21
0.999	7586.2	1585.1	1901.7	0.21

in microseconds. The tandem lands below the bound at every quantile, with the ratio in the last column between 0.14 and 0.21 against the bound’s 0.25; the slack is the ρ/N utilisation effect described in the remark after Theorem 1. Under the Poisson null at the same message rate ($\rho = 0.40$, which is far above the corpus utilisation and is used here only to make the Poisson wait visible) the single-stage wait quantiles at $q = 0.9, 0.99, 0.999$ are 10.3, 24.8, 38.5 μs and the tandem’s are 5.1, 7.5, 9.2 μs , again inside the bound; at the corpus utilisation both collapse to 0 and $3h$ respectively, which is Corollary 2.

C Numerical check of the unequal-stage claims

The same generator as Appendix B (Borel(0.9) clusters, $\rho = 0.40$, $T = 10 \mu\text{s}$, $h = 1.7 \mu\text{s}$), with each cluster’s members spread uniformly over $3 \mu\text{s}$ instead of placed as an instantaneous batch and 4×10^5 messages, run through the partitions of Remark 2.2 with an explicit per-stage FIFO recursion. The last two columns are the tail-excess ratio $(\Delta_{99}(P) - \text{hops} \cdot h)/\Delta_{99}(\text{single})$ and the bound $s_{\max}/T$ from Theorem 1.

partition	$s_{\max}$	hops	p_{50}	p_{99}	$p_{99.9}$	ratio	bound
10	10.0	0	388.2	5719.0	8249.3	1.00	1.00
7 + 3	7.0	1	213.6	3429.5	5297.1	0.60	0.70
3 + 7	7.0	1	213.6	3429.5	5297.1	0.60	0.70
5 + 5	5.0	1	134.9	2265.5	3574.2	0.39	0.50
7 + 2 + 1	7.0	2	215.3	3431.2	5298.8	0.60	0.70
4 + 3 + 3	4.0	2	104.6	1746.5	2714.6	0.30	0.40
4 + 3 + 2 + 1	4.0	3	106.3	1748.2	2716.3	0.30	0.40
2.5×4	2.5	3	66.4	1049.8	1562.2	0.18	0.25

Latencies are end-to-end in microseconds. The three claims of Remark 2.2 read off directly: the tail excess depends on the partition only through $s_{\max}$ and sits below the bound at every row; $7 + 3$ and $3 + 7$ are identical to the reported precision; and $7 + 2 + 1$ differs from $7 + 3$, and $4 + 3 + 2 + 1$ from $4 + 3 + 3$, by exactly one hop cost $h = 1.7 \mu\text{s}$ at every quantile. The $3 \mu\text{s}$ intra-cluster spread, which is above $s_{\max}$ for the last row, is why the equal-stage ratio here (0.18) sits slightly below the batch-input value of Appendix B (0.21): a cluster that arrives spread out lets the faster tandem drain part of it before the next member lands.